



Characterization of turbulent pseudo-boiling in transcritical and doubly-transcritical oxygen–methane flames

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ABSTRACT

This study investigates the turbulent pseudo-boiling process in transcritical and doubly transcritical LOx-methane flames using large-eddy simulations (LES) under conditions relevant to next-generation liquid rocket engines (LRE). The definition of the turbulent pseudo-boiling rate, based on the concept of displacement speed, is introduced for reacting flows and LES frameworks. A dedicated strategy is developed to identify the effective pseudo-boiling state when using turbulent combustion models, where the actual laminar pseudo-boiling typically occurs at the sub-grid scale. The thermodynamic structure of the pseudo-boiling transition is analyzed in mixture fraction space using laminar flamelet calculations and LES results. Although the pseudo-phase transition of the propellants occurs almost as a pure fluid, the choice of mixing rules for thermodynamic property evaluation is shown to influence the thermal breakup of the jet and the resulting flame morphologies. The LES data are then used, in conjunction with the turbulent pseudo-boiling rate definition, to investigate statistically the heating of the propellants. The analysis reveals that the mass transfer under transcritical conditions is mainly driven by the normal diffusion fluxes induced by the non-premixed flame between the two propellant streams. We found that LOx pseudo-boils at similar rates in both transcritical and doubly-transcritical flames. In contrast, the rates of cryogenic methane in the doubly-transcritical case are significantly lower, not only because of the inherently higher methane thermal inertia but also due to the occurrence of a non-isotropic insulating region influenced by the interplay between thermodynamic conditions and turbulent motions.

1. Introduction

In recent years, oxygen–methane has emerged as a promising propellant combination in space propulsion due to its numerous advantages, especially when compared to more traditional combinations such as oxygen–kerosene and oxygen–hydrogen. The combined effects of higher specific impulse, better cooling capabilities [1], and lower likelihood of coking and soot formation [1,2] make methane a competitive alternative to heavier hydrocarbons such as RP-1, particularly in the context of reusable liquid rocket engines (LRE). Conversely, although oxygen–methane combustion delivers a lower specific impulse than its oxygen–hydrogen counterpart, the benefits of using methane as a fuel over hydrogen derive from its higher liquid storage density and boiling point—those characteristics lead to lighter, smaller and simpler cryogenic storage and thermal handling systems [1,3].

LRE thrust chambers typically operate under supercritical pressure conditions to achieve high efficiency and compact geometric size. In recent decades, the scientific community has made considerable efforts to understand the phenomena of injection, mixing, and combustion

processes at supercritical regimes. Experimental investigations targeting LRE-like conditions for both inert and reacting jets [4–8], reveal that when the operating pressure exceeds the propellant critical values, the classical subcritical breakup mechanism into ligaments and droplets ceases. Instead, the jets disintegrate through a turbulent process, giving rise to a diffuse interface with a steep but finite mixing gradient. These findings have led to the development of distinct modeling paradigms. Instead of considering the injection process as a Lagrangian spray of liquid droplets, the current standard approach is to treat the supercritical diffuse interface as an Eulerian continuum [9,10], which in turn requires real-fluid thermodynamic models. In this regard, the well-established approach assumes that the fluids behave as a single-phase mixture. The deviation from the ideal gas state is tackled with cubic [11,12] or multiparametric equations of state (EoS) in conjunction with mixing rules [13,14].

Cryogenic propellant injection provides an additional thermodynamic distinction that characterizes LRE combustors, leading to the so-called *transcritical* regime [15]. When heated, a transcritical pure fluid

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Nomenclature**Roman letters**

A	Area
d	Diameter
C_x	Model constant
c_p	Mixture isobaric specific heat
D	Mixture fraction diffusivity
E	Mixture ratio
h	Mixture enthalpy
$\dot{m}$	Mass flow rate
M	Molecular weight
N_a	Avogadro number
N_s	Number of species
p	Pressure
$P(\cdot)$	Probability density function
R	Universal gas constant
$S_{i,j}$	Components of the strain rate tensor
Sc	Schmidt number
t	Time
T	Temperature
u_i	Components of the velocity vector
v	Specific volume
v_Z	Mixture fraction displacement speed
VR	Velocity ratio
Z	Mixture fraction
Z'^2	Mixture fraction variance
$\bar{Z}$	Compressibility factor
Y	Mass fraction
X	Mole fraction
x_j	Cartesian coordinates
J	Momentum flux ratio
$\mathcal{J}$	Mass flux per unit area

Greek letters

δ_{ij}	Components of the Kronecker delta
Δ	Characteristic filter width
ϵ	Lennard-Jones potential well depth
μ	Dynamic viscosity
λ	Thermal conductivity
ρ	Density
σ	Lennard-Jones collision diameter
τ_{ij}	Components of the stress tensor
χ	Scalar dissipation rate
ψ	Generic thermodynamic variable
ω	Acentric factor
$\dot{\omega}$	Chemical source term

Sub- and Superscripts

$\bar{\cdot}$	Filtered quantity
$\bar{\cdot}^*$	Density-weighted filtered quantity
c	Critical point property
k	General index of the species
n	Normalized property
pb	Pseudo-boiling point property
sgs	Sub-grid-scale property
st	Stoichiometric point property

Abbreviations

CFD	Computational fluid dynamics
DNS	Direct numerical simulation
DTF	Doubly transcritical flame
EoS	Equation of state
LOx	Liquid oxygen
LES	Large-eddy Simulations
LRE	Liquid rocket engine
MFM	Multi fluid mixing
SGS	Sub-grid scale
SFM	Single fluid mixing
TF	Transcritical flame

In aerospace propulsion applications, the unique characteristics of supercritical fluids have been primarily studied for two key applications. The first concerns heat transfer processes with solid walls, where phenomena such as heat transfer deterioration can significantly affect the thermal performance of cooling systems [23–31]. The second, which is the focus of the present work, regards the complex interaction of thermodynamic nonlinearities due to the pseudo-boiling mechanism and turbulent flow associated with the injection, mixing, and combustion processes under LRE conditions.

On this latter aspect, transcritical nitrogen injection experiments [32,33] confirmed the importance of the physical properties of the fluid in the thermal breakup of the jet. When the fluid has to cross the c_p peak, an extended liquid core develops, highlighting the significant influence of real-fluid effects when injecting in a liquid-like state. Regarding reacting flows, experimental studies of liquid oxygen (LOx) flames using hydrogen [34,35] or methane [36] as fuels have shown that the near-critical enhancement of thermodynamic and transport properties plays a crucial role in the combustion process and flame morphologies. Specifically, their analysis suggests that in transcritical conditions, heat and mass transfer processes between the dense inner core and the surrounding lighter gas stream are driven by turbulent mixing. Thus, pseudo-vaporization of cryogenic oxygen becomes the rate-limiting process.

Despite the remarkable findings, limited data sets can be extracted from experimental diagnostics under LRE-like operating conditions due to the severe thermodynamic regimes. In this context, numerical tools such as computational fluid dynamics (CFD) are becoming of paramount importance to investigate fundamental physical aspects hindered by experimental investigations. Over the past few decades, non-premixed combustion at supercritical pressure has been largely studied by leveraging the counterflow diffusion flame setting [37–40] or as a direct solution of the flamelet equations in mixture fraction space [14,41,42]. The impact of several fundamental aspects, such as strain rate, inlet temperature, and pressure on flame structure, have been analyzed. Regarding the LOx pseudo-boiling phenomena, their results suggest that prominent real-fluid effects are present during the transition from a supercritical liquid-like to gas-like, involving abrupt property variations. These effects are restricted to a small temperature range, and the transition from a dense to an ideal gas occurs almost as a pure fluid. However, additional considerations exploring the laminar pseudo-boiling mechanism of liquid-like methane (LCH₄) remain largely unexplored.

High-fidelity computations, such as direct numerical simulations (DNS) aimed at studying the pseudo-boiling process, have mainly been used to investigate non-reacting cryogenic mixing [43–48]. They all pointed out the strong influence between the mass transfer and the pseudo-boiling mechanism. The steep gradient stratification induced by the specific heat capacity's peculiar distribution delays the jet's complete mixing. On the other hand, DNS featuring a reacting oxygen–methane mixture has been used to characterize flame root stabilization mechanism at the lips of a coaxial injector, under doubly-transcritical

or mixtures undergoes the *pseudo-boiling* process [16,17]. As a result of this pseudo-phase transition, distinct supercritical thermodynamic states with liquid-like and gas-like thermophysical properties can be delineated [16,18–21]. The supercritical extension of the vapor–liquid coexistence line, also known as the Widom line, usually identified by the locus of isobaric-specific heat c_p maxima [16,22], divides these states.

LOx-LCH₄ conditions [49], i.e., where both propellants are injected in a dense cryogenic state. For similar thermodynamic operating points and in the case of a splitter plate configuration to investigate flame structure [50], and flame propagation following the ignition [51]. Nevertheless, simplified two-dimensional configurations have been studied, given the considerable computational cost associated with the high-resolution requirements and detailed modeling of kinetic mechanisms.

In this context, the large-eddy simulation (LES) approach is the main compromise when dealing with laboratory or industrial-scale devices to investigate complex phenomena at a reasonable cost. In the last decade, LES dealing with transcritical liquid-oxygen-gaseous-methane LOx-GCH₄ flames has been performed by few research groups [52–57]. Most of these efforts have focused on preventing spurious pressure oscillations, a common problem when dealing with highly stratified conditions such as those encountered in LREs [58,59]. Recently, LES has been performed to study, under the doubly-transcritical regime, the heat-release dynamics when the fuel inflow is subjected to acoustic forcing [60]. While substantial progress has been made in understanding transcritical combustion, LES studies addressing the pseudo-boiling mechanism are still lacking, further motivating the present study.

The primary objective of the present work is the characterization of the turbulent pseudo-boiling mechanism in oxygen–methane cryogenic flames. Specifically, we provide a detailed analysis of this mechanism in the mixture fraction space. We have studied the effects of two different mixing models, taken from the literature, in both laminar and turbulent conditions on the pseudo-phase transition. Leveraging a consistent real-fluid flamelet-based LES numerical framework for the simulation of high-pressure turbulent flows in LRE combustion chambers that can account for pseudo-boiling occurring at the sub-grid-scale (SGS) level, we perform Large-Eddy simulations of transcritical and doubly-transcritical LOx-methane flames at near-critical pressure. We have qualitatively and quantitatively discussed the flame morphologies and characteristics of these two different thermodynamic injection conditions. In particular, the doubly-transcritical case is of specific relevance for the community since only a few studies in the existing literature have dealt with this peculiar case. We provide a consistent definition of the turbulent pseudo-boiling rate for reacting mixture within an LES framework, extending the work of Lapenna [17]. Leveraging the ensuing data, we investigate the statistical characteristics of the mass transfer rate in pseudo-boiling conditions and its interplay within a turbulent flow.

The paper is organized as follows: Section 2 reports on the numerical techniques employed, including real-fluid thermodynamic and transport modeling, with particular details on the multi-component mixing models and their implementation in a flamelet-based framework, together with the governing equations. Section 3 describes the definition of the turbulent pseudo-boiling rate within an LES framework based on the concept of the displacement speed of the mixture fraction. Section 4 presents the test cases chosen to investigate the pseudo-boiling mechanism, detailing the thermodynamic injection conditions, the numerical framework, and the simulation setup employed. In addition, the grid sensitivity analysis and the evaluation of the LES resolution quality are reported. Section 5 presents the results. Specifically, we analyze the pseudo-boiling mechanism in the mixture fraction space, accompanied by a discussion of the main flame characteristics of the two cases studied. Finally, we report a statistical analysis of the turbulent pseudo-boiling rate. Conclusive remarks of the work are reported in Section 6.

2. Methods

2.1. Real-fluid thermodynamics and transport modeling

The ideal gas law does not adequately capture the interplay among pressure, density, and temperature at cryogenic and supercritical pressure conditions. For this reason, numerical simulations of transcritical

flows require appropriate thermodynamic closure models. In CFD studies, thermodynamic models based on a cubic EoS are widely used to account for real gas effects due to their acceptable accuracy, straightforward formulation, and reliability for a wide range of fluids. In the present work, we use the cubic Soave–Redlich–Kwong (SRK) EoS [12]:

$$p(T, \rho) = \frac{\rho RT}{M - b\rho} - \frac{a\alpha(T)\rho^2}{M^2 + 2Mb\rho}, \quad (1)$$

where p is the pressure, R is the universal gas constant, ρ is the density of the mixture, T is the temperature and M is the molar mass. The non-ideal mixture parameters a and b account for the attractive and repulsive forces between the molecules. The parameter $\alpha(T)$ is a correction factor that considers the species polarity. The single-component attractive and repulsive parameters, assuming a chemically stable k th species, can be evaluated as follows:

$$a_k \alpha_k(T) = 0.427 \frac{(RT_{c,k})^2}{p_{c,k}} \left[1 + c_k \left(1 - \sqrt{\frac{T}{T_{c,k}}} \right) \right]^2, \quad (2)$$

$$b_k = 0.08664 \frac{RT_{c,k}}{p_{c,k}}, \quad (3)$$

$$c_k = 0.48508 + 1.55171\omega_k - 0.15613\omega_k^2. \quad (4)$$

In the above expressions, $T_{c,k}$, $p_{c,k}$, and ω_k are the critical temperature, pressure, and acentric factor of k th species, respectively. Thus, the non-ideal parameters of chemically stable species such as CH₄, O₂, H₂, H₂O, CO₂, CO, or metastable species such as H₂O₂, can be determined from their critical state conditions and acentric factors. However, since chemically unstable species such as radicals recombine, it is impossible to define the states of pure species and their corresponding critical points. Following [61,62], one way to overcome this problem is to assume that the k th chemically unstable species is a Lennard-Jones gas. In turn, critical properties can be estimated [61], and analogous expressions for the attractive and repulsive parameters read as

$$a_k \alpha_k(T) = 5.55 N_a^2 \epsilon_k \sigma_k^3 \left[1 + 0.485 \left(1 - \sqrt{\frac{T}{T_{c,k}}} \right) \right]^2, \quad (5)$$

$$b_k = 0.85551 N_a \sigma_k^3, \quad (6)$$

where N_a is the Avogadro number, ϵ_k and σ_k the Lennard-Jones potential and molecular diameter of the k th species, respectively. In addition, in the same way as done in [61,62], for these types of molecules and those not available in the National Institute of Standards and Technology (NIST) database [63], we considered the acentric factor ω_k null. A summary of the critical properties and acentric factors for all species considered in the present work is reported in Table 1.

Thermodynamic properties such as enthalpy h and isobaric specific heat c_p are estimated using the departure function formalism [13,14]. Thus, they can be evaluated as the sum of the ideal gas counterparts, computed using the 7-coefficient NASA7 polynomials [64], and a departure function to account for real fluid effects, analytically specialized for the SRK EoS [14]. Accurate modeling is also required to evaluate molecular transport properties at supercritical pressures. We used the well-established semi-empirical method of Chung et al. [65] to calculate the mixture viscosity μ and thermal conductivity λ .

Fig. 1 shows the density, isobaric specific heat, viscosity, and thermal conductivity of the pure propellant species O₂ and CH₄ considered in this work, evaluated with the reported models and compared with the NIST [63] reference data. Both thermodynamic and transport properties are in overall good agreement compared with the NIST database.

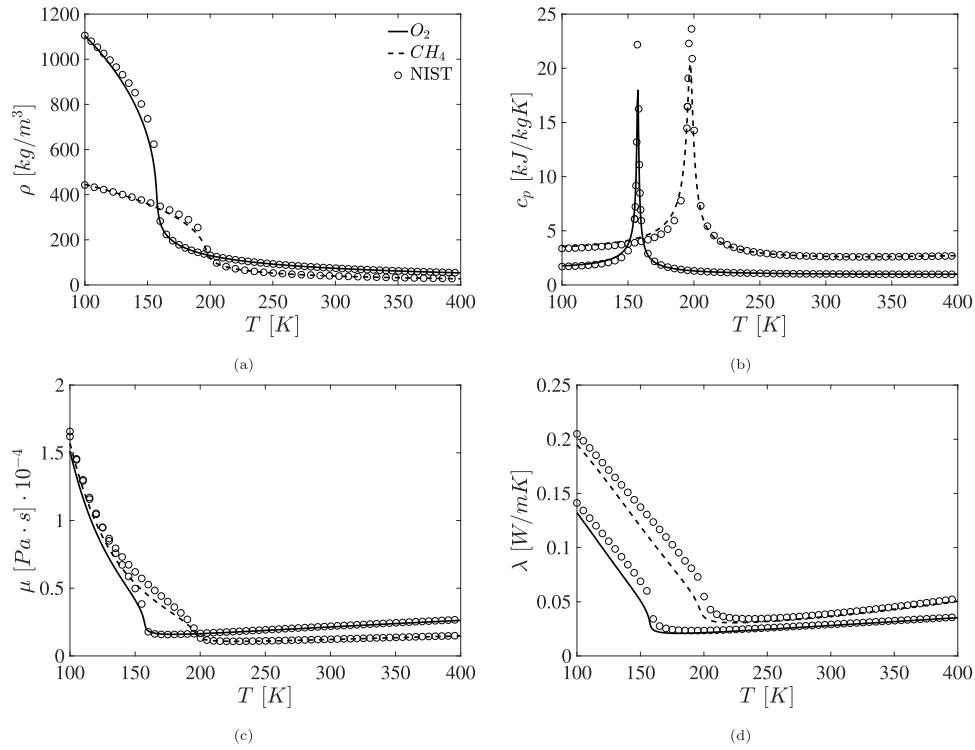


Fig. 1. Comparison of thermodynamic and transport properties evaluated with the SRK-EoS and Chung method for oxygen (solid lines) and methane (dashed lines) at 56 bar with the NIST [63] database (dots), of: (a) density, (b) isobaric specific heat, (c) dynamic viscosity, and (d) thermal conductivity.

Table 1

Critical properties T_c , p_c , v_c , and acentric factor ω for all species considered in this work.

Species	T_c [K]	p_c [bar]	ω	v_c [cm³/mol]
H	190.82	310.13	0.0000	17.069
H ₂	33.180	13.000	-0.216	64.935
C	93.962	35.450	0.0000	71.072
CH	105.28	68.511	0.0000	41.204
CH ₂	189.50	46.739	0.0000	108.71
CH ₃	189.50	46.739	0.0000	108.71
O	105.28	70.880	0.0000	41.204
CH ₄	190.60	46.100	0.0110	98.600
OH	105.28	70.880	0.0000	41.204
H ₂ O	647.10	220.64	0.3440	55.865
CO	134.45	34.987	0.0497	90.090
C ₂ H	275.04	54.009	0.0000	136.55
C ₂ H ₂	308.30	61.380	0.1900	112.00
C ₂ H ₃	275.04	54.009	0.0000	136.55
C ₂ H ₄	282.50	50.600	0.0866	131.00
HCO	655.37	191.67	0.0000	91.670
C ₂ H ₅	332.02	56.439	0.0000	157.74
C ₂ H ₆	305.32	49.000	0.0990	147.00
CH ₂ O	655.37	191.70	0.0000	91.670
CH ₃ O	548.77	147.82	0.0000	99.546
CH ₂ OH	548.77	147.82	0.0000	99.546
O ₂	154.58	50.430	0.0222	73.529
CH ₃ OH	634.05	179.99	0.0000	94.456
HO ₂	141.34	47.860	0.0000	81.926
H ₂ O ₂	141.34	47.860	0.0000	81.926
HCCO	197.40	170.98	0.0000	30.957
CH ₂ CO	573.77	124.10	0.0000	123.97
HCCOH	573.77	124.10	0.0000	123.97
CO ₂	304.18	73.800	0.2239	94.339
CH ₃ O ₂	634.05	180.10	0.0000	94.456
CH ₃ O ₂ H	634.05	180.10	0.0000	94.456
C ₂ H ₅ O	619.31	97.726	0.0000	169.92
C ₂ H ₅ O ₂	619.31	97.726	0.0000	169.92
C ₂ H ₅ O ₂ H	619.31	97.726	0.0000	169.92

2.1.1. Mixing models

As discussed in Section 1, we employ two mixing models to extend the SRK-EoS to a multicomponent mixture. The first widely adopted approach to determine mixture thermodynamic properties of supercritical fluids assumes that the properties of a single-phase mixture are equivalent to those of a reference hypothetical fluid [13,66]. The commonly used nomenclature to classify these methods is single or one-fluid models [67]. Hereafter, we will refer to the latter as the single-fluid-mixing (SFM) model. The mixture attractive and repulsive terms in Eq. (1) are obtained through a set of non-linear mixing rules [13,14]:

$$a\alpha(T) = \sum_{k=1}^{N_s} \sum_{j=1}^{N_s} X_k X_j a_{kj} \alpha_{kj}(T), \quad (7)$$

$$b = \sum_{k=1}^{N_s} X_k b_k, \quad (8)$$

where N_s indicates the number of species, X the molar fraction. To obtain the off-diagonal terms $a_{kj} \alpha_{kj}$ from the critical properties, we employed the following pseudo-critical combination rules [13,14]:

$$T_{c,kj} = \sqrt{T_{c,k} T_{c,j}} (1 - \mathcal{K}_{kj}), \quad (9)$$

$$p_{c,kj} = \mathcal{Z}_{c,kj} (RT_{c,kj} / v_{c,kj}),$$

$$\omega_{kj} = \frac{1}{2} (\omega_k + \omega_j),$$

with

$$v_{c,kj} = \frac{1}{8} (v_{c,k}^{1/3} + v_{c,j}^{1/3}), \quad \mathcal{Z}_{c,kj} = \frac{1}{2} (\mathcal{Z}_{c,k} + \mathcal{Z}_{c,j}), \quad (10)$$

where $\mathcal{Z}_c$ and v_c are the compressibility factor and molar volume at the critical state, respectively. The binary interaction parameter $\mathcal{K}_{kj}$ is available from experiments only for a limited number of binary mixtures and thus is often assumed to be zero [14]. To evaluate the mixture transport properties μ and λ , we use the mixing rules reported in the work of Chung et al. [65]. Despite the widespread use of this method, in addition to the assumption of the existence of the mixture in

single-phase states and its expression via a hypothetical reference fluid, which should be similar to the involved fluids, there are limits to the conditions of applicability of the SFM model. In particular, the strongest assumption implies that the critical parameters of all fluids involved in the mixture should have similar values within a factor of two, which is not usually met for both LOx-H₂ and LOx-CH₄ combustion (see Table 1). For additional details, the entire set of assumptions is discussed in [66] and rigorously reviewed in [67].

Therefore, even if the SFM found successful and largely application for LRE coaxial injection [50–52,55–59,68,69], to avoid the mentioned limitations and to explore the effects of other approaches, we employed the multi-fluid-mixing (MFM) model developed by Banuti et al. [67]. This method was originally developed for LOx-GH₂ combustion, but its assumptions hold in the case of LOx-GCH₄ combustion [55], and is based on the physical characteristics of cryogenic injection and non-premixed flames. Both experimental [4] and numerical [37,39,40,68] studies show that mixing in LOx-GH₂ non-premixed flames occurs solely among ideal gases. Real gas effects are only present within the cold LOx core; the transition from liquid-like to gas-like states occurs in pure oxygen. Therefore, mixing within the hot fuel-rich region and the flame zone is an ideal gas process. Thus, with the MFM model, ideal and real gas modeling can be decoupled. Only the properties of the fluids expected to exhibit non-ideal behavior (e.g., pure species propellants) are evaluated using a real gas EoS, and all other species are considered ideal gases while mixing is modeled as an ideal process. Following [67], the generic mixture thermodynamic variable ψ is computed using the mass-weighted formula as

$$\psi = \frac{\sum_k^{N_s} \rho_k \psi_k}{\sum_i^{N_s} \rho_k}, \quad (11)$$

where ρ_k and ψ_k are the partial densities and individual species properties, respectively. Regarding the transport properties, we use the Wilke [70] rule to evaluate the mixture viscosity μ , while we estimate the mixture thermal conductivity using the mixture-averaged formulation [71].

Note that since we are investigating both transcritical LOx-GCH₄ and doubly transcritical LOx-LCH₄ flames, we have considered real-fluid effects for both propellant species, i.e., O₂ and CH₄. While the MFM assumptions are valid for the former injection conditions, the latter, to the best of the author's knowledge, were never addressed. Therefore, in Section 5.1 we discuss the mixing model effects on the pseudo-boiling mechanism and transition. Moreover, we use the SRK-EoS and the Chung method to tackle real-fluid effects for the selected non-ideal species for two reasons. In the first place, as shown in Fig. 1, they exhibit acceptable accuracy. Secondly, to circumscribe the sole effect of the mixing model, the use of advanced multiparametric EoS, or NIST data, could lead to an unfair comparison with the SFM model.

2.2. Combustion model

In this study, we employ the flamelet [72] approach for the combustion model. This concept covers regimes where chemical time scales are shorter than the turbulence time scales, e.g., in conditions that occur in most practical combustion systems, particularly in LRE high-pressure combustors [8,37–39]. The local flame structures embedded in a turbulent-non-premixed flame are described as an ensemble of laminar flames, known as flamelets, by solving the flamelet equations in the mixture fraction space Z , given below under the unitary Lewis number assumption and in their unsteady form:

$$\frac{\partial Y_k}{\partial t} = \frac{\chi}{2} \frac{\partial^2 Y_k}{\partial Z^2} + \frac{\dot{\omega}_k}{\rho} \quad k = 1, \dots, N_s, \quad (12)$$

$$\frac{\partial T}{\partial t} = \frac{\chi}{2} \frac{1}{c_p} \left[\frac{\partial^2 h}{\partial Z^2} - \sum_{k=1}^{N_s} h_k \frac{\partial^2 Y_k}{\partial Z^2} - \frac{2}{\rho \chi} \sum_{k=1}^{N_s} h_k \dot{\omega}_k \right], \quad (13)$$

where Y_k and $\dot{\omega}_k$ are the mass fractions and chemical source term of species k . The scalar dissipation rate χ , which represents the local non-equilibrium effects, is assumed to be known apriori using a one-parametric distribution [72]:

$$\chi(Z) = \chi_{st} \exp \left[-2 \left(\operatorname{erfc}^{-1}(2Z) \right)^2 \right], \quad (14)$$

where the subscript st denotes the stoichiometric condition. The flamelet manifold, hence the steady-state solutions of Eqs. (12)–(13) are solved with the OpenSMOKE++ [73] library modified in-house to deal with flows under LRE relevant conditions [74] using the thermodynamics models presented in the previous section. To model the chemical reaction rates, we employed the detailed kinetic mechanism RAMEC [75], which is tailored for high-pressure methane oxidation and comprises 38 species (see Table 1) and 190 reactions.

Fig. 2 shows an example of a real-fluid oxygen–methane flame structure in terms of temperature and density as a function of the mixture fraction Z . Specifically, Fig. 2(a) reports the results for a near-equilibrium flames ($\chi_{st} = 1 \text{ s}^{-1}$) at transcritical conditions, i.e. pressure $p = 56 \text{ bar}$, oxidizer $T_{O_2} = 85 \text{ K}$ and fuel inlet temperature $T_{CH_4} = 288 \text{ K}$, respectively. Firstly, we verify the profiles against the results reported in the work of Kim et al. [69], where the same EoS and SFM models were employed. As can be observed, the temperature profiles are essentially superimposed between the models. On the other hand, it is worth mentioning that for the density, slight differences are appreciable in a small range on the oxidizer side, with the MFM providing higher values. At the same pressure and scalar dissipation rate, Fig. 2(b) shows the flame structure of doubly-transcritical conditions, i.e., the fuel temperature is set to $T_{CH_4} = 120 \text{ K}$, from which similar considerations to the transcritical case can be extracted.

In contrast, for the dynamic viscosity and thermal conductivity displayed in Fig. 3, the estimate of μ with MFM is moderately higher than with SFM in the range of $0.02 < Z < 0.4$, while for both sides of the pseudo-boiling region their values are practically identical. Similarly, the thermal conductivity λ is insensitive to the mixing models in the cryogenic regions. However, it shows large differences in the range of $0.02 < Z < 0.9$, with the MFM model giving higher estimates. Due to the focus of the present work, a detailed analysis of the impact of the employed mixing models on the isobaric-specific heat is reported in Section 5.1.

One of the key issues for the simulation of transcritical flows is the non-linear relationship between temperature, pressure, and mixture composition introduced by a real-fluid EoS. Its application in the LES context gives rise to a resolved part and a SGS residual term due to the filtering procedure [74,76]. While the magnitude of the SGS contribution in the filtered EoS remains rather limited with ideal gas flows [46,74,76], where a no-model approximation is well suited, i.e., the SGS term is considered null, in the context of LES of transcritical flames modeling of the EoS is indeed required. Flamelet-like models provide a robust platform for modeling the unresolved part of the EoS using presumed probability density function (PDF) methods, especially when an EoS independent low-Mach ($M < 0.3$) number formulation is used [46,74]. With this modeling strategy, the laminar thermochemical manifold, which takes the form $\psi = \psi(Z, \chi_{st}; p_0)$, where ψ is the generic thermodynamic property, and p_0 the thermodynamic pressure, can be efficiently pre-computed and stored in look-up tables. Then, to model the unresolved turbulent flames, filtered values of all thermodynamic and transport properties, are obtained through a bi-variate PDF [77]:

$$\tilde{\psi} = \int_0^\infty \int_0^1 \psi(Z, \chi_{st}; p_0) P(Z, \chi_{st}) dZ d\chi_{st}, \quad (15)$$

assuming statistical independence between Z and χ_{st} , the joint PDF $P(Z, \chi_{st})$ can be approximated by $P(Z) P(\chi_{st})$. Thus, turbulence-chemistry interactions are modeled using a presumed (β -PDF) for the SGS fluctuations of the mixture fraction using $\tilde{Z}$ and its variance $\tilde{Z}''^2$ as inputs.

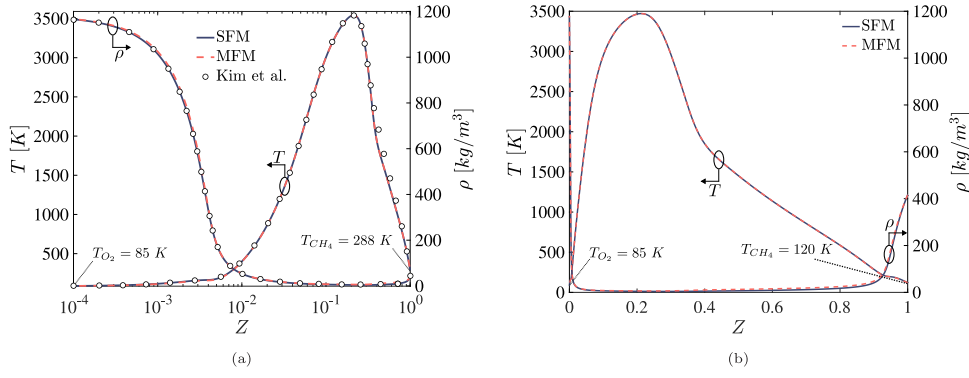


Fig. 2. Comparison of a near-equilibrium laminar flame structure in terms of temperature and density using the SFM (blue-lines) and MFM (red-lines) model, for (a) transcritical flame and (b) doubly-transcritical flame.

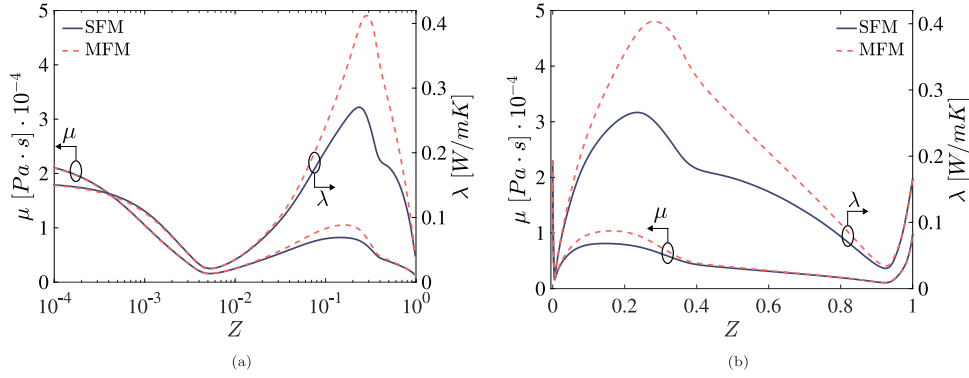


Fig. 3. Comparison of a near-equilibrium laminar flame structure in terms of dynamic viscosity and thermal conductivity using the SFM (blue-lines) and MFM (red-lines) model, for (a) transcritical flame and (b) doubly-transcritical flame.

Conversely, a Dirac-delta distribution is used for the scalar dissipation rate. As a result, the unresolved contribution to all the thermodynamic properties is inherently contained in the thermo-chemical manifold, which takes the form:

$$\tilde{\psi} = \tilde{\psi}(\tilde{Z}, \tilde{Z}''^2, \tilde{\chi}_{st}; p_0) \quad (16)$$

this procedure ensures a consistent real-fluid EoS modeling as discussed in [46,74]. For all the cases reported in the rest of the manuscript, the flamelet tables are discretized using 1024, 32, and 20 points w.r.t the access parameters.

2.3. Governing equations

The Favre-filtered conservation equations of mass and momentum used to formulate the LES approach are

$$\frac{\partial \bar{\rho}}{\partial t} + \frac{\partial \bar{\rho} \tilde{u}_j}{\partial x_j} = 0, \quad (17)$$

$$\frac{\partial \bar{\rho} \tilde{u}_i}{\partial t} + \frac{\partial \bar{\rho} \tilde{u}_i \tilde{u}_j}{\partial x_j} = -\frac{\partial \bar{p}_1}{\partial x_i} + \frac{\partial \bar{\tau}_{ij}}{\partial x_j} + \frac{\partial \tau_{ij}^{sgs}}{\partial x_j}, \quad (18)$$

where $\bar{\cdot}$ and $\tilde{\cdot}$ respectively denote the spatial filtering and the Favre-filtering operation, i.e., $\tilde{\cdot} = \bar{\rho} \tilde{\cdot} / \bar{\rho}$, while u_i are the velocity components. In the momentum equation, $\bar{p}_1$ is the hydrodynamic pressure, an $\mathcal{O}(M^2)$ term of the thermodynamic pressure p_0 , which is uniform over the computational domain, and constant in time since open domains are considered in this study. The superscript *sgs* stands for the modeled SGS contribution from the filtering procedure. The fluid mixture is assumed to be Newtonian, with the resolved and modeled viscous stress tensors written as

$$\bar{\tau}_{ij} + \tau_{ij}^{sgs} = 2(\bar{\mu} + \bar{\rho} v^{sgs}) \left(\tilde{S}_{ij} - \frac{1}{3} \delta_{ij} \tilde{S}_{kk} \right), \quad (19)$$

where S_{ij} is the strain rate tensor, and δ_{ij} is the Kronecker-Delta. The SGS kinematic viscosity v^{sgs} is modeled using the WALE model [78].

In flamelet-based LES approaches for variable-density turbulent flows, two additional transport equations are solved for the filtered mixture fraction and its variance to describe the turbulent flame with the look-up process of the flamelet libraries [79]. These equations read:

$$\frac{\partial \bar{\rho} \tilde{Z}}{\partial t} + \frac{\partial \bar{\rho} \tilde{u}_j \tilde{Z}}{\partial x_j} = \frac{\partial}{\partial x_j} \left[\left(\bar{D} + D^{sgs} \right) \frac{\partial \tilde{Z}}{\partial x_j} \right], \quad (20)$$

$$\begin{aligned} \frac{\partial \bar{\rho} \tilde{Z}''^2}{\partial t} + \frac{\partial \bar{\rho} \tilde{u}_j \tilde{Z}''^2}{\partial x_j} = & \frac{\partial}{\partial x_j} \left[\left(\bar{D} + D^{sgs} \right) \frac{\partial \tilde{Z}''^2}{\partial x_j} \right] \\ & - 2\bar{\rho} \tilde{\chi} + 2 \left(\bar{D} + D^{sgs} \right) \left(\frac{\partial \tilde{Z}}{\partial x_j} \right)^2, \end{aligned} \quad (21)$$

where $\bar{D} = (\lambda/c_p)$ and $D^{sgs} = \bar{\rho} v^{sgs} / Sc_t$ are the molecular and sub-grid diffusivity, respectively. The turbulent Schmidt number is $Sc_t = 0.7$ in all simulations. Following [80], the scalar dissipation rate $\tilde{\chi}$ is modeled decomposing it into resolved and SGS term as

$$2\bar{\rho} \tilde{\chi} = 2\bar{D} \left(\frac{\partial \tilde{Z}}{\partial x_j} \right)^2 + C_\chi D^{sgs} \frac{\tilde{Z}''^2}{\Delta}, \quad (22)$$

where C_χ is a model constant set to 2 as done in [55,57,79,81], and Δ denotes the local filter size computed as the cube root of the cell volume.

3. Turbulent pseudo-boiling rate concept

As discussed in the introduction, the rate at which a cryogenic fluid pseudo-boils is crucial during propellant injection in LREs. The localized high thermal inertia during the mixture pseudo-phase transition delays the jet thermal breakup, influencing the dynamic of mass

transfer across propellant shear layers and ultimately affecting the mixing and combustion processes. Following [17], to characterize the interplay between the pseudo-boiling mechanism in a turbulent flow, we propose the concept of the mixture fraction's displacement speed, which represents the velocity of a mixture fraction iso-surface relative to the convective velocity that occurs via diffusion in the normal direction to the surface. Using this concept, the mass flux per unit area across a surface in pseudo-boiling conditions, namely the pseudo-boiling rate, is obtained. This approach has been applied extensively to various combustion problems for premixed [82,83] and non-premixed flames [84,85]. The displacement speed of the filtered mixture fraction is

$$v_{\tilde{Z}} = \mp \frac{\nabla \cdot \left[\left(\bar{D} + D^{sgs} \right) \nabla \tilde{Z} \right]}{\bar{\rho} |\nabla \tilde{Z}|} \hat{n}, \quad \hat{n} = \pm \frac{\nabla \tilde{Z}}{|\nabla \tilde{Z}|}, \quad (23)$$

where $\hat{n}$ is the unit normal vector. As discussed earlier, we also investigate the pseudo-boiling rate under doubly transcritical conditions, i.e., for the stream of both oxidizer and fuel. Thus, the different sign in Eq. (23) is due to the need to define a different convention to establish a common direction of fluid convection for the two cases. To help visualize this definition, Fig. 4 shows a sketch representation of a turbulent, non-premixed flame (lower panel) and its depiction in mixture fraction space (upper panel). The red solid line represents the flame surface identified with the location of the stoichiometric mixture fraction $\tilde{Z}_{st}$. The dashed blue and green lines denote the hypothetical pseudo-boiling mixture fraction surfaces $\tilde{Z}_{pb}$, for the LOx and LCH₄, respectively. For both, the unit normal vector points toward the flame position. For the former (LOx), the normal is oriented toward the increasing mixture fraction, while the latter (LCH₄) is toward the decreasing mixture fraction. Given this definition, for both propellant streams, if $v_{\tilde{Z}} > 0$, the mass flux in pseudo-boiling conditions is directed toward the flame, while for $v_{\tilde{Z}} < 0$ is directed toward their respective pure streams.

Furthermore, following Echekki and Chen [86], the diffusive term of Eq. (23) can be recast into a tangential and normal component:

$$v_{\tilde{Z}} = \underbrace{\mp \frac{(\bar{D} + D^{sgs})}{\bar{\rho}} \nabla \cdot \hat{n}}_{v_{\tilde{Z}}^{CRV}} \mp \underbrace{\frac{1}{\bar{\rho} |\nabla \tilde{Z}|} \frac{\partial}{\partial \eta} \left[(\bar{D} + D^{sgs}) \frac{\partial \tilde{Z}}{\partial \eta} \right]}_{v_{\tilde{Z}}^{ND}}, \quad (24)$$

where η is a scalar coordinate normal to the iso-surface, this expansion indicates that $v_{\tilde{Z}}$ has two contributions. The first, denoted as $v_{\tilde{Z}}^{CRV}$, is associated with the local curvature of the surface, locally measured by the term $\nabla \cdot \hat{n}$. The second contribution $v_{\tilde{Z}}^{ND}$, measures the motion of the iso-surface relative to the flow in the normal direction due to diffusion.

At this stage, we can define the mass flux per unit area across the iso-surface as $J = \mp \bar{\rho} v_{\tilde{Z}}$. Furthermore, by conditioning this definition on a specific scalar value of the mixture fraction, i.e., the pseudo-boiling state $\tilde{Z} = \tilde{Z}_{pb}$, we can define the turbulent pseudo-boiling rate as

$$J_{pb} = \mp (\bar{\rho} v_{\tilde{Z}}) |_{\tilde{Z}_{pb}} = \mp \left(J_{pb}^{CRV} + J_{pb}^{ND} \right), \quad (25)$$

the two terms J_{pb}^{CRV} and J_{pb}^{ND} are the curvature and normal diffusion components, respectively, of the turbulent pseudo-boiling rate, resulting from the expansion of Eq. (24). Moreover, to investigate the actual role of the diffusive transport mechanisms acting across the pseudo-boiling interface, the total flux J_{pb} is decomposed into its laminar (or resolved) J_{pb}^{lam} and SGS J_{pb}^{sgs} contributions as

$$J_{pb} = \mp \left[\frac{\nabla \cdot (\bar{D} \nabla \tilde{Z})}{|\nabla \tilde{Z}|} + \frac{\nabla \cdot (D^{sgs} \nabla \tilde{Z})}{|\nabla \tilde{Z}|} \right] \bigg|_{\tilde{Z}_{pb}} = \mp \left(J_{pb}^{lam} + J_{pb}^{sgs} \right) \bigg|_{\tilde{Z}_{pb}}, \quad (26)$$

this decomposition allows a more detailed understanding of how molecular and turbulent diffusion quantitatively affect the pseudo-boiling mechanism.

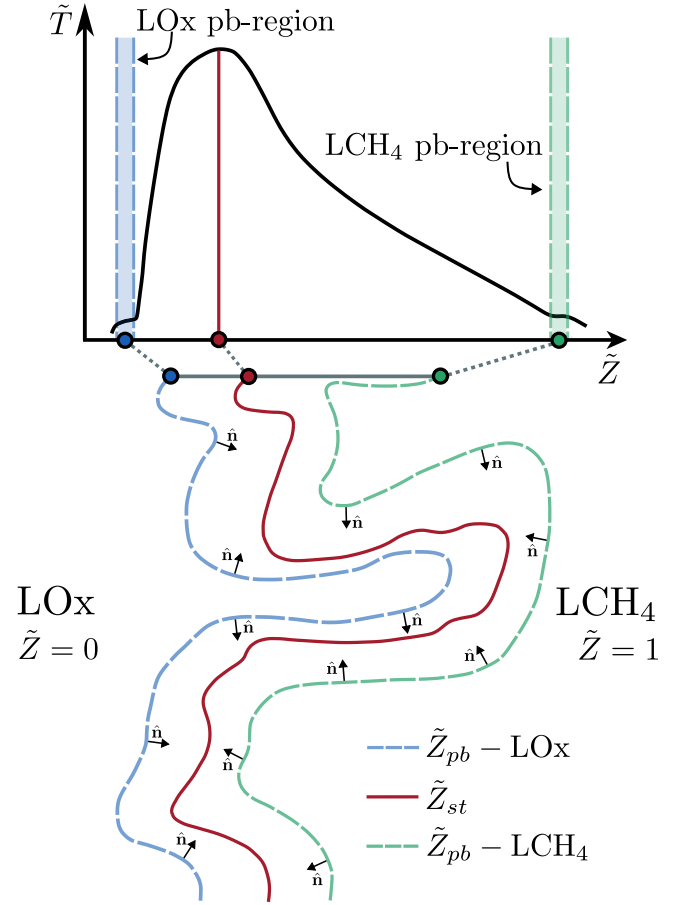


Fig. 4. Schematic representation of a turbulent non-premixed flame (lower panel) and its depiction in mixture fraction space (upper panel). The red solid line represents the stoichiometric surface $\tilde{Z}_{st}$, while the dashed blue and green lines denote the hypothetical pseudo-boiling mixture fraction surfaces $\tilde{Z}_{pb}$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Note that in [17], the concept of pseudo-boiling rate was applied to a single species mixture and in a DNS framework. Therefore, once the pseudo-boiling state is assigned, the iso-surface is characterized by a single value of thermodynamic properties. In contrast, the present work addresses a complex problem by considering a reacting mixture with LES-SGS terms and filtered thermodynamic properties due to unresolved mixture fraction fluctuations. For these reasons, the selection of a proper scalar iso-surface to investigate the turbulent pseudo-boiling is carefully addressed in Section 5.1. In addition, the presented framework can also be leveraged in combustion-LES approaches employing different flamelet-based thermo-chemical manifolds or finite-rate chemistry.

4. Test case and numerical setup

In the present study, we investigated the turbulent pseudo-boiling rate by adapting two reference thermodynamic injection states from the experimental tests described in Singla et al. [36], conducted at the Mascotte cryogenic combustion facility. The Mascotte combustion chamber is a $50 \times 50 \text{ mm}^2$ square duct with a length of 400 mm and a convergent-divergent nozzle. The propellants are delivered by a single coaxial shear injector, with oxygen injected from an inner circular stream and methane coaxially into the combustor. In particular, Singla et al. [36] performed several tests varying the operating chamber pressure, from subcritical to supercritical conditions, and the methane

Table 2

Operating and injection conditions of the considered transcritical and doubly transcritical flames.

p [bar]	T_{O_2} [K]	T_{CH_4} [K]	ρ_{O_2} [kg/m ³]	ρ_{CH_4} [kg/m ³]	$\dot{m}_{O_2}$ [g/s]	$\dot{m}_{CH_4}$ [g/s]	u_{O_2} [m/s]	u_{CH_4} [m/s]	E	J	VR
TF											
56.1	85	288	1180	42	44.4	143.1	3.70	63.5	0.31	39	33
DTF											
56.1	85	120	1180	414	44.4	143.1	3.70	20.2	0.31	39	10.5

Table 3

Detailed injector dimensions for the test cases considered.

Case	d_1 [mm]	d_2 [mm]	d_3 [mm]	d_4 [mm]
TF	3.6	5.0	5.6	10.0
DTF	3.6	5.0	5.6	7.29

injection temperature, resulting in transcritical and doubly-transcritical flames.

In the first place, we simulated a LOx-GCH₄ transcritical flame (TF) under the thermodynamic operating conditions of the G2 test case described in [36]. The chamber pressure is 56.1 bar, which is supercritical for fuel and oxidizer. Oxygen is injected in a liquid-like state at $T = 85$ K, while the supercritical methane injection temperature is 288 K. Under such conditions, the density of oxygen ρ_{O_2} and methane ρ_{CH_4} are 1180 kg/m³ and 42 kg/m³, respectively. The mass flow rates $\dot{m}$ of liquid oxygen and gaseous methane are 44.4 g/s and 143.1 g/s, respectively, resulting in a mixture ratio $E = \dot{m}_{O_2}/\dot{m}_{CH_4} = 0.31$. Several research groups have numerically studied this test case, as it is considered a relevant benchmark for the validation of numerical solvers [52,57] and turbulent combustion models [55,56,74] under LRE conditions. Regarding injector details, we have based our work on injector dimensions such as those reported in [69,81,87–90]. Specifically, the liquid oxygen is supplied through a diverging duct with an inlet diameter of 3.6 mm and an outlet diameter of 5 mm. The LOx post-tip width is 0.3 mm, while the methane coaxial annulus has a height of 2.2 mm. The resulting LOx and methane injection velocities are 3.70 m/s and 63.5 m/s, respectively. Given these conditions, the other important parameters controlling the flame length in coaxial jets are the momentum flux ratio $J = (\rho_{CH_4} u_{CH_4}^2)/(\rho_{O_2} u_{O_2}^2)$ and the velocity ratio $VR = u_{CH_4}/u_{O_2}$, which are close to 39 and 33, respectively.

Secondly, we simulated a LOx-LCH₄ doubly-transcritical flame (DTF); for this test case, we keep the same operating pressure, mass flow rates (hence the same mixture ratio), and liquid oxygen injection temperature as the transcritical flame. To obtain the doubly-transcritical condition, we considered the methane injection temperature in a liquid-like state at $T = 120$ K, which yields a density of 414 kg/m³. In addition, the DTF case shares the same inner round injector dimensions and post-tip width as the TF case. On the other hand, to have a similar injector parameters design, we imposed the same momentum flux ratio of the TF case, resulting in a methane injection velocity of 20.2 m/s, whereas, given the mass flow rate, the methane annulus height is 0.845 mm for the DTF condition. A summary of the operating and injection conditions of the considered cases is reported in Table 2.

We perform the LES with an in-house finite-volume tabulated chemistry solver based on the OpenFOAM [91] and OpenSMOKE++ [73] suites. This framework has already been applied to high-pressure reacting and non-reacting flows under LRE-like conditions in a large number of previous works based on both LES [92,93] and Unsteady Reynolds-Averaged Navier–Stokes (uRANS) [74,94–96] approach. It was recently validated against highly stratified transcritical conditions [97] as encountered in this work. To mitigate the spurious pressure oscillations induced by the real-fluid and to improve the numerical stability, we adopted the solution strategy first proposed by Jarczyk and Pfitzner [98] and extensively used in OpenFOAM pressure-based

solvers [56,99,100]. Specifically, the pressure correction equation is adapted for real-fluid flows using a Taylor expansion to linearize the cubic EoS. The detailed derivation procedure can be found in [98]. The second change, w.r.t the ideal-gas approach, regards the PIMPLE algorithm [101]. In particular, the thermodynamic properties in the trans-supercritical regime exhibit strong nonlinearities and, as a consequence, the transport equations are strongly coupled. Following [98], the scalars equations and the evaluation of the thermodynamic properties were incorporated into the PISO loop [101] along with the pressure-correction equation. Diffusive and convective fluxes are estimated using a second-order accurate centered difference scheme. Temporal integration is performed using the implicit Euler scheme, with a maximum Courant–Friedrichs–Lewy condition of $CFL = 0.4$. After flushing the initial condition and reaching a fully developed flow field, we collected the temporal statistics for all simulations reported in the paper for $\Delta t = 60$ ms, which correspond to $\Delta t \approx 140 \cdot d_{O_2}/u_{conv}$ and $\Delta t \approx 105 \cdot d_{O_2}/u_{conv}$ for the TF and DTF, respectively, with u_{conv} being the convective velocity evaluated using the expression given in [102].

Fig. 5 shows a schematic representation of the simulation domain, while Table 3 reports details of the injector dimensions. To reduce the computational cost, we consider a numerical domain length of 200 mm in the axial direction, which has been tested to be sufficiently large to avoid backflow effects and interactions between the flame and the outlet section. All walls are modeled as adiabatic with standard wall-law boundary conditions, while a Dirichlet pressure boundary condition is applied to the outlet plane. Turbulent velocity fluctuations with an intensity of 3% are superimposed to the bulk flow at the injection of both propellants with the synthetic eddy method [103]. LES were performed using an O-type hexahedral grid. We evaluated the sensitivity of the results to the grid resolution with a systematic mesh refinement analysis and three different grids, designated M1, M2, and M3, containing approximately 6.3, 13.1, and 23.7 million finite volumes, respectively. The smallest cells are located at the injector lip with a width of 150, 100, and 75 μ m for the three meshes. From the injection plate to downstream, where the gradients are less steep, the grids become progressively coarser in the axial direction with a cell-to-cell expansion ratio of less than 1.003. The simulations were performed on Intel Xeon 6230 Gold processors with 60, 120, and 240 cores for the three grids. The total computational cost, including the initial transient to reach a statistically steady state, and the time required to collect temporal statistics, was approximately 13,000, 26,000, and 78,000 CPU hours, respectively.

Fig. 6 shows the grid sensitivity analysis for the TF case, where we compare the time-averaged radial profiles of the mean $\langle \bar{T} \rangle$ (top panel) and root-mean-square $\langle \bar{T}' \rangle$ (bottom panel) at three axial positions, namely $x = 15$ mm, $x = 45$ mm and $x = 70$ mm. Close to the injector, at $x = 15$ mm, the behavior of the flame opening angle, identified by the temperature peak, is captured with all grid resolutions, and slight differences are observed. However, within the recirculation region, marked by the temperature plateau from $y \approx 10$ mm to the chamber wall, the profiles for M2 and M3 grids are closer than for M1. The resolution impact is evident further downstream, at $x = 45$ mm, the temperature with the coarse mesh at the centerline ($y = 0$ mm) is significantly higher, while the profiles of M2 and M3 overlap, indicating that the coarse grid underestimates the flame length. Also, at $x = 70$ mm, where the flow is homogeneous, the temperature predicted with

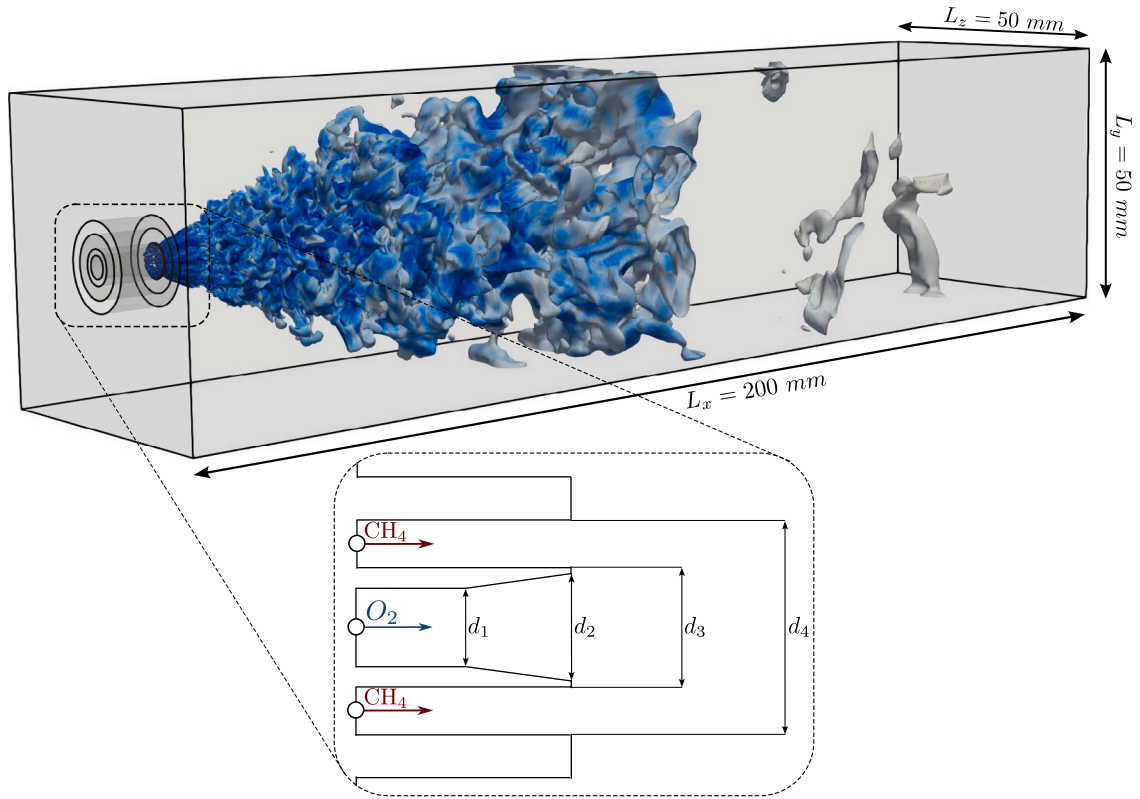


Fig. 5. Sketch representation of the entire computational domain (upper panel), inlets and injector (bottom panel).

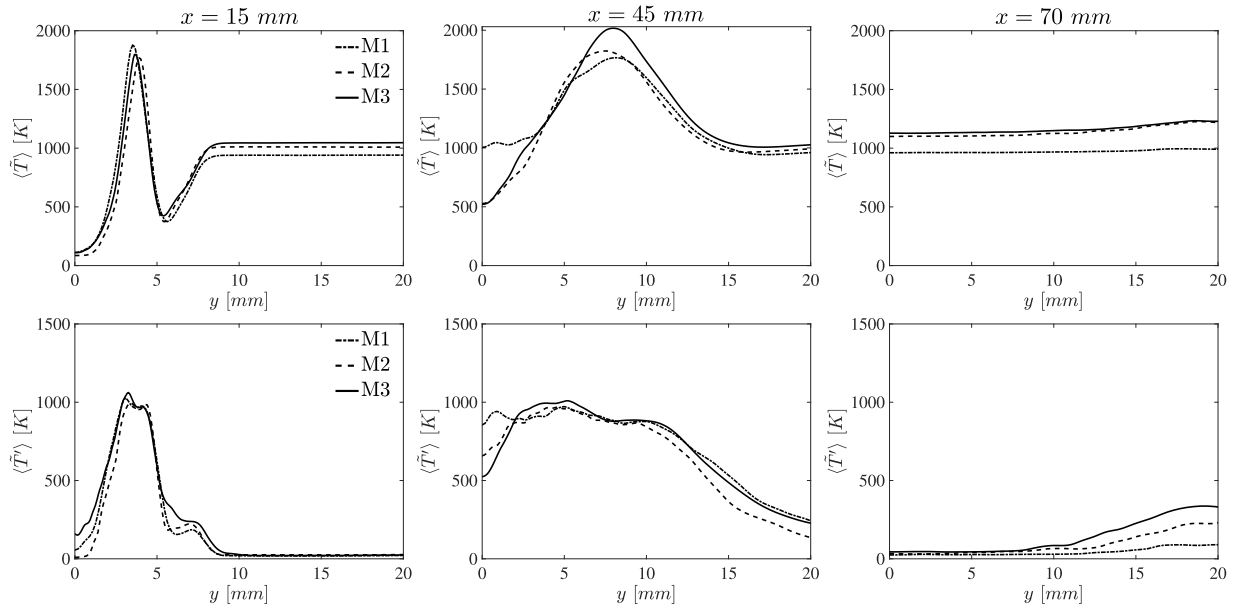


Fig. 6. Grid sensitivity analysis for the TF case. Radial temporal mean (first row) and root mean square (second row) temperature profiles at $x = 15$ mm, $x = 45$ mm, and $x = 70$ mm from the injector.

M2 and M3 is in close agreement, while M1 shows a slightly lower value.

Despite the higher computational cost of the M3 grid, this resolution is employed throughout the paper. This is done to increase the fidelity of the turbulent pseudo-boiling analysis reported in the results. To further assess the LES resolution quality of the selected grid, Fig. 7 shows the spatial distribution of the time average of Celik's LES quality

index LESIQ [104] (upper half of the figure) and the local filter size Δ (lower half of the figure), for both simulated cases. The LESIQ presents rather high values ($\text{LESIQ} > 0.85$) in the flame region (see next section for spatial topology) and over the whole domain, confirming an indicator of good resolution (i.e., verification) and thus the appropriate LES representation of both cases.

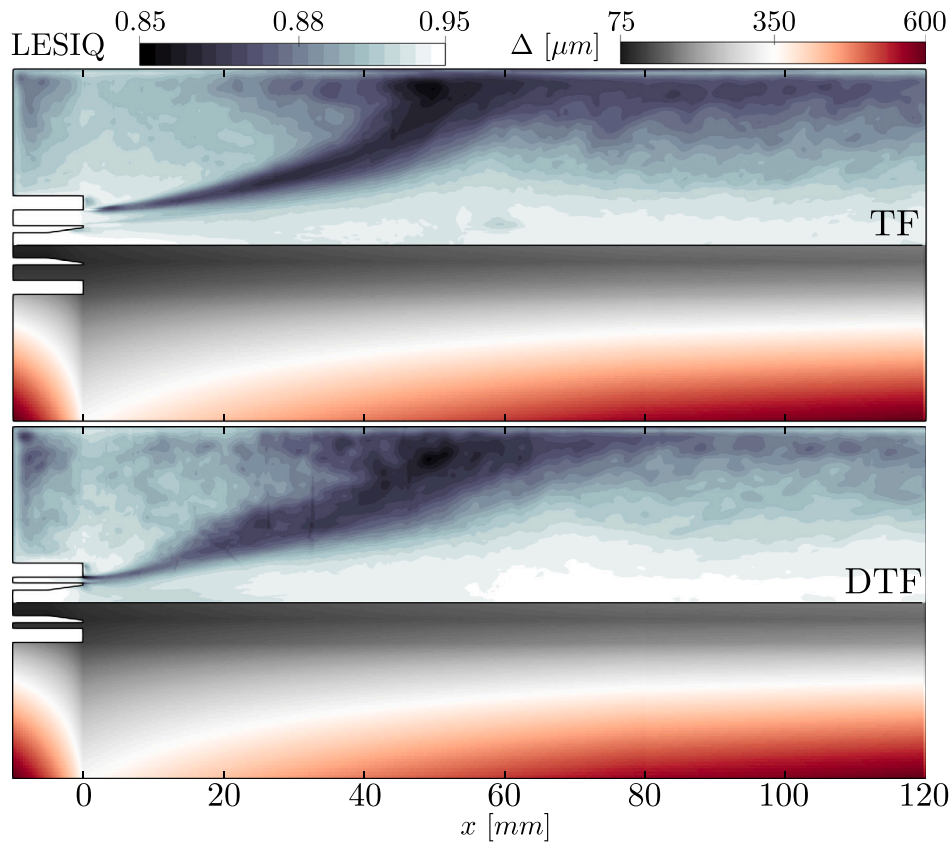


Fig. 7. Contour maps of mean LESIQ quality index by Celik [104] (upper half), and the local filter size Δ (lower half). Top panel: TF case. Bottom panel: DTF case.

5. Results and discussion

5.1. Pseudo-boiling analysis in mixture fraction space

The pseudo-boiling phenomenon is interpreted as an isobaric pseudo-phase transition between a supercritical liquid-like and a gas-like state. For a single component mixture, it can be quantified in a continuous sense as a transcritical process that starts from a subcritical temperature $T < T_{pb}$ and ends at supercritical temperature $T > T_{pb}$, where the pseudo-boiling temperature T_{pb} denotes the temperature corresponding to the maximum specific heat at constant pressure. On the other hand, for multicomponent mixtures, a validated theoretical model providing quantification of the pseudo-boiling loci as a function of mixture composition is lacking in the current literature. Nevertheless, from a thermodynamic point of view, it can be identified as the c_p local maximum. Still, the accuracy of the prediction depends on the employed EoS and mixing rules. Thus, in the present section, we first investigate the structure of the isobaric heat capacity in the mixture fraction space resulting from the laminar flamelets solution, with the employed mixing models and given the thermodynamic baseline conditions of Table 2. In this regard, the left panel of Fig. 8 compares the $c_p(Z)$ profiles of a near-equilibrium flame $\chi_{st} = 1 \text{ s}^{-1}$ and for a reference strain level $\chi_{st} = 1000 \text{ s}^{-1}$. Note that we choose this value for comparison since the maximum scalar dissipation rates obtained from the LESs are less than $\bar{\chi}_{st} \approx 10^4 \text{ s}^{-1}$, which is far from the extinction limit $\chi_{st}^{ext} \approx 6 \cdot 10^5 \text{ s}^{-1}$ at the investigated pressure (see also [55]).

For the oxidizer-rich side shown in Fig. 8(a), contrary to what has been observed for the flame structures in Fig. 2, the influence of the mixing model on the c_p behavior, and therefore the pseudo-boiling process, is pronounced. The value of the mixture fraction corresponding to the peak of c_p is for both models $Z_{pb} \approx 0.0038$, which corresponds to a pseudo-boiling temperature $T_{pb} \approx 157 \text{ K}$. The MFM model, which only considers the effect of the non-ideal fluid on the propellants, as

expected, gives a lower maximum value than the SFM and is approximately equal to the pseudo-boiling value of pure oxygen $(c_p)_{pb} \approx 18 \text{ kJ/kg K}$ (see Fig. 1(b)). While the value obtained with the SFM model is considerably higher $(c_p)_{pb} \approx 31.5 \text{ kJ/kg K}$. This difference is attributable to the effect of composition on the departure from ideality when using the SFM model. In particular, the non-negligible mass fractions composition at the pseudo-boiling mixture fraction is $(Y_i)_{pb} \approx Y_{O_2} + Y_{CO_2} + Y_{H_2O} = 0.98 + 0.0105 + 0.0085$. Although previous studies for both LOx-hydrogen [39,40] and LOx-methane combustion [42] have discussed that the pseudo-boiling transition occurs under nearly pure oxygen conditions, it is important to underline that the small oxygen impurities due to the presence of CO_2 and H_2O have a significant impact. Even if these impurities have a mass fraction of $\approx 2\%$, they are characterized by high critical properties (see Table 1), greatly influencing the prediction of c_p when using the SFM model. The thermal inertia variations associated with the c_p peak have been recognized in [39,40,105] as a primary factor that impedes the heating of the LOx core by creating a thermal barrier. Thus, the differences observed with the mixing models are expected to play a role in the heating process of liquid oxygen and, in turn, affect the pseudo-boiling rate.

Similar observations hold for the fuel-rich side shown in Fig. 8(b). The values of the pseudo-boiling mixture fraction are very close for both models and lie in a small range between $Z_{pb} \approx 0.94$ – 0.95 , which corresponds to a pseudo-boiling temperature comprised in $T_{pb} = 195$ – 197 K . The estimates of the c_p peaks differ less between the two models used than on the LOx-side, and for the near-equilibrium flame ($\chi_{st} = 1 \text{ s}^{-1}$), they are $(c_p)_{pb} \approx 18 \text{ kJ/kg K}$ and $(c_p)_{pb} \approx 22.3 \text{ kJ/kg K}$ for MFM and SFM model, respectively. Moreover, in contrast to the LOx-side, where the profiles are substantially superimposed, the effects of non-equilibrium conditions ($\chi_{st} = 1000 \text{ s}^{-1}$) on the flame structure are observed. The profiles move towards pure-fuel conditions as the scalar dissipation rate increases, where it is interesting to note the increase

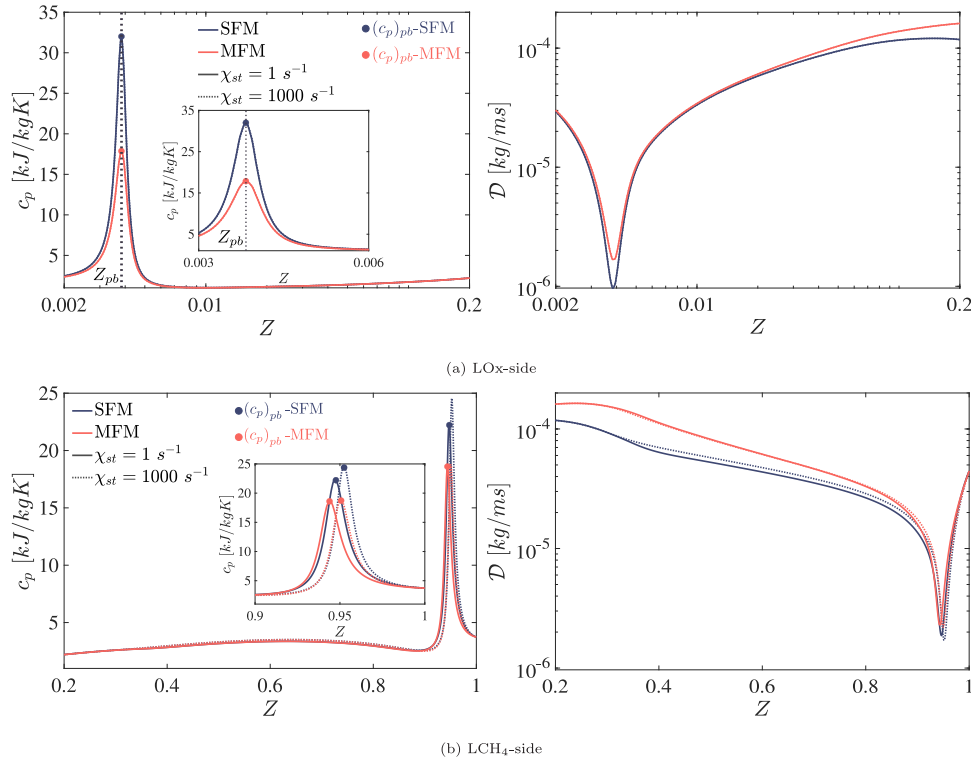


Fig. 8. Laminar pseudo-boiling analysis. Isobaric specific heat profiles (left-panel) and thermal diffusivity (right-panel) as a function of the mixture fraction Z with $\chi_{st} = 1 \text{ s}^{-1}$ (solid-lines) and $\chi_{st} = 1000 \text{ s}^{-1}$ (dotted-lines), using the SFM (blue-lines) and MFM (red-lines) model, for (a) LOx-side and (b) LCH₄-side.

in the peak of c_p using the SFM model due to mixture composition variation.

Moreover, a further important difference w.r.t the oxidizer-rich side concerns the composition during the pseudo-boiling transition, if in the former case, it occurs under nearly pure oxygen conditions, in contrast for the latter, the composition for the near-equilibrium flame is $(Y_i)_{pb} \approx Y_{H_2} + Y_{H_2O} + Y_{CH_4} + Y_{CO} + Y_{CO_2} = 0.0056 + 0.0207 + 0.9155 + 0.0484 + 0.0088$. Since H_2 and CO are expected to be in gas-like conditions at these temperatures (see Table 1), the species that lead to the differences in the mixing models are still the water and the carbon dioxide.

Lastly, the effects of the mixing models on the thermal diffusivity $D = \lambda/c_p$, shown in the right panel of Fig. 8, are caused mainly by the differences in isobaric-specific heat within both sides of the pseudo-boiling region, since the thermal conductivity estimates are close (see Fig. 3). On the other hand, in the regions where the real-fluid effects are moderate and the c_p estimates overlap, the differences can be attributed to those observed in the thermal conductivity. As a consequence, except for the pure (or nearly pure) propellant states, the thermal diffusivity estimated by the MFM model is always higher than that of the SFM in the whole mixture fraction space. Note that the percentage deviations reach 63%–73% on the cryogenic sides, and are in the range of 20%–80% in $0.2 < Z < 0.9$.

After the laminar analysis, we now investigate the effect in the pseudo-boiling process of the turbulence-chemistry interaction, modeled using the presumed β -PDF for the SGS fluctuations of the mixture fraction. Moreover, to investigate the turbulent pseudo-boiling rate, we illustrate a criterion to select a proper pseudo-boiling surface in terms of mixture fraction $\tilde{Z}$ within a turbulent LES framework.

Fig. 9 compares a time instant of the scattered LES data of filtered $\bar{c}_p(\tilde{Z}, \tilde{Z}''^2)$ in mixture fraction space, where each dot is color-coded by its normalized SGS mixture fraction variance $\tilde{Z}''^2 = \tilde{Z}''^2 / (\tilde{Z}(1 - \tilde{Z}))$, against nominal $\bar{c}_p$ profiles obtained with three value of $\tilde{Z}''^2$, namely $\tilde{Z}''^2 = 0$ (i.e. the laminar solution), $\tilde{Z}''^2 = 0.005$ and $\tilde{Z}''^2 = 0.05$. The left panel of Fig. 9(a) shows the results for the LOx-side of the

transcritical flame obtained with the SFM model, while the right panel shows the results obtained with the MFM one. Note that, similarly to the laminar analysis, for brevity, we report only the results of the TF since they are qualitative and quantitatively similar to the oxidizer-rich side of the DTF. The results show that for both mixing models, most of the scatter data are enclosed within the $\tilde{Z}''^2 = 0.005$ and $\tilde{Z}''^2 = 0.05$ solutions, thus even with the finest grid, the SGS mixture fraction variance in the cryogenic region is not negligible, and leads to a pronounced decrease of the maximum values of $\bar{c}_p$, and a shifting toward higher mixture fraction values w.r.t the laminar solution. This comparison also highlights the differences in how the mixing model handles turbulence-chemistry interaction and SGS variance effects. As observed for the laminar solution (see Fig. 8), the maximum values of the scatter data obtained with the MFM model are lower than the SFM one. The differences can be quantified by observing the solid blue (SFM) and red (MFM) line, which shows the employed criterion to select the pseudo-boiling iso-surface, i.e., is identified as the local maximum of the conditional mean $\langle \bar{c}_p | \tilde{Z} \rangle$ of the LES data. Based on this definition, this criterion is expected to represent, in an average sense, the location of an *effective turbulent pseudo-boiling* process in LES. In particular, for the selected time-instant, the turbulent pseudo-boiling specific heat predicted with the SFM is $(\bar{c}_p)_{pb} \approx 2.9 \text{ kJ/kg K}$, while with MFM is $(\bar{c}_p)_{pb} \approx 2.4 \text{ kJ/kg K}$, which allows to identified the mixture fraction value that resembling pseudo-boiling conditions, that is for both models $\tilde{Z}_{pb} \approx 0.008$. Note that both values are significantly different from the laminar (or no-model) values previously reported.

The same analysis is displayed in Fig. 9(b) for LCH₄-side of the doubly-transcritical flame, in this case, even if the scatter data are still enclosed within the $\tilde{Z}''^2 = 0.005$ and $\tilde{Z}''^2 = 0.05$ solutions, the smearing effect due to presumed β -PDF recorded for the LOx-side is less pronounced. For low variance level the isobaric specific heat reaches values up to 11 kJ/kg K, while based on the mentioned criterion, the turbulent pseudo-boiling specific heat predicted with the SFM is $(\bar{c}_p)_{pb} \approx 8 \text{ kJ/kg K}$, while with MFM is $(\bar{c}_p)_{pb} \approx 7.4 \text{ kJ/kg K}$, which corresponds to a pseudo-boiling mixture fraction $\tilde{Z}_{pb} \approx 0.95$.

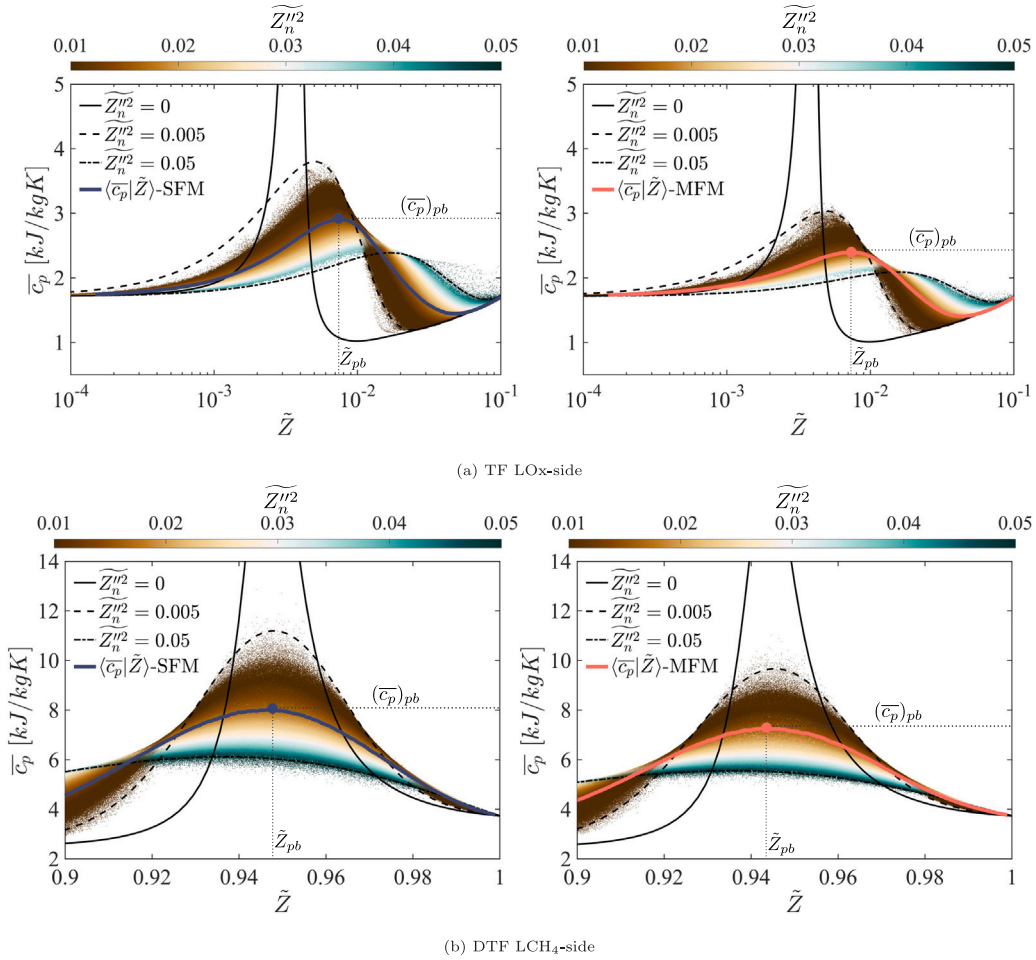


Fig. 9. Turbulent pseudo-boiling analysis. Instantaneous $\bar{c}_p(\tilde{Z}, \tilde{Z}_n'^2)$ LES scatter data color-coded with the normalized mixture fraction variance $\tilde{Z}_n'^2$ using the SFM (left-panel) and MFM (right panel), compared with a-priori $\bar{c}_p$ profiles obtained with three values of $\tilde{Z}_n'^2$, i.e., $\tilde{Z}_n'^2 = 0$ (solid), $\tilde{Z}_n'^2 = 0.005$ (dashed) and $\tilde{Z}_n'^2 = 0.05$ (dashed-dotted), for (a) TF LOx-side and (b) DTF LCH₄-side. The blue (SFM) and red (MFM) solid lines denote the conditional mean $\langle \bar{c}_p | \tilde{Z} \rangle$ of the LES data. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

It is important to note that using a no-model approximation for the thermochemical properties would lead to a similar identification of a pseudo-boiling surface in both laminar and turbulent analyses. This could result in an incorrect interpretation of the pseudo-boiling occurring at the SGS level. For further details on this aspect, the reader should refer to [74,76].

5.2. Flowfields overview

In this section, we discuss qualitatively the characteristics of the flame patterns of the two cases investigated. Fig. 10 shows different instantaneous fields through a longitudinal section of the computational domain. The upper panel of the figures shows the results of the TF, while the lower panel shows the DTF. For the sake of brevity, and since the differences between the mixing models are more appreciable with the averaged fields, Fig. 10 reports the results with the SFM model.

From the injection and up to $x \approx 40$ mm, the flame topology of the TF shown in Fig. 10(a) exhibits an elongated, thin-layer pattern with a limited spreading angle. The flame boundary is sharply defined near the injector, with prominent turbulent eddies confined to the high shear between the two jets. Further downstream, large-scale turbulent structures develop and gradually spread outward, while high-temperature regions abruptly end at about $x \approx 60$ mm. This observation can also be seen in the density field illustrated in Fig. 10(b). The high-density core is narrow and well-defined up to $x \approx 40$ mm. Beyond this point, pockets of liquid-like oxygen begin to detach, and the core gradually diffuses as

it moves downstream until it is completely vaporized. The white isoline of the stoichiometric mixture fraction $\tilde{Z}_{st} = 0.2$ represents the position of the resolved flame. To further characterize the flame morphology, Fig. 10(c) shows the axial velocity field and the white zero velocity iso-lines delineating the recirculation zones. The central core pseudo-boiling triggers an intense expansion of the oxidizer flow, imparting outward momentum to the surrounding flame and redirecting it toward the chamber wall, resulting in lateral recirculation zones. As reported in previous LES results [55–57], a large central recirculation zone can be observed at the end of the reaction zone.

Conversely, the temperature field of the DTF shows a higher flame wrinkling near the injection region and a slightly longer flame length than the TF. Qualitative differences are also significant in the rear part of the flame, a broader and thickened distribution is observed where large scales dominate and turbulence decreases. Additionally, the injection of cryogenic methane results in significantly lower temperatures in the corner recirculation zones compared to the transcritical case. Further downstream, the doubly-transcritical conditions lead to a less homogeneous flow, indicating a reduced mixing efficiency. The unique characteristics of cryogenic methane injection can be seen in the density field. The liquid-like coaxial jet extends significantly within the chamber, reaching up to $x \approx 35$ mm, so the methane stream is in a gas-like state downstream of this location. Regarding the axial velocity, the DTF shows smaller scales and a more irregular distribution w.r.t the TF, suggesting that the dynamics induced by the two thermodynamic injections significantly impact the turbulent mixing process. The central

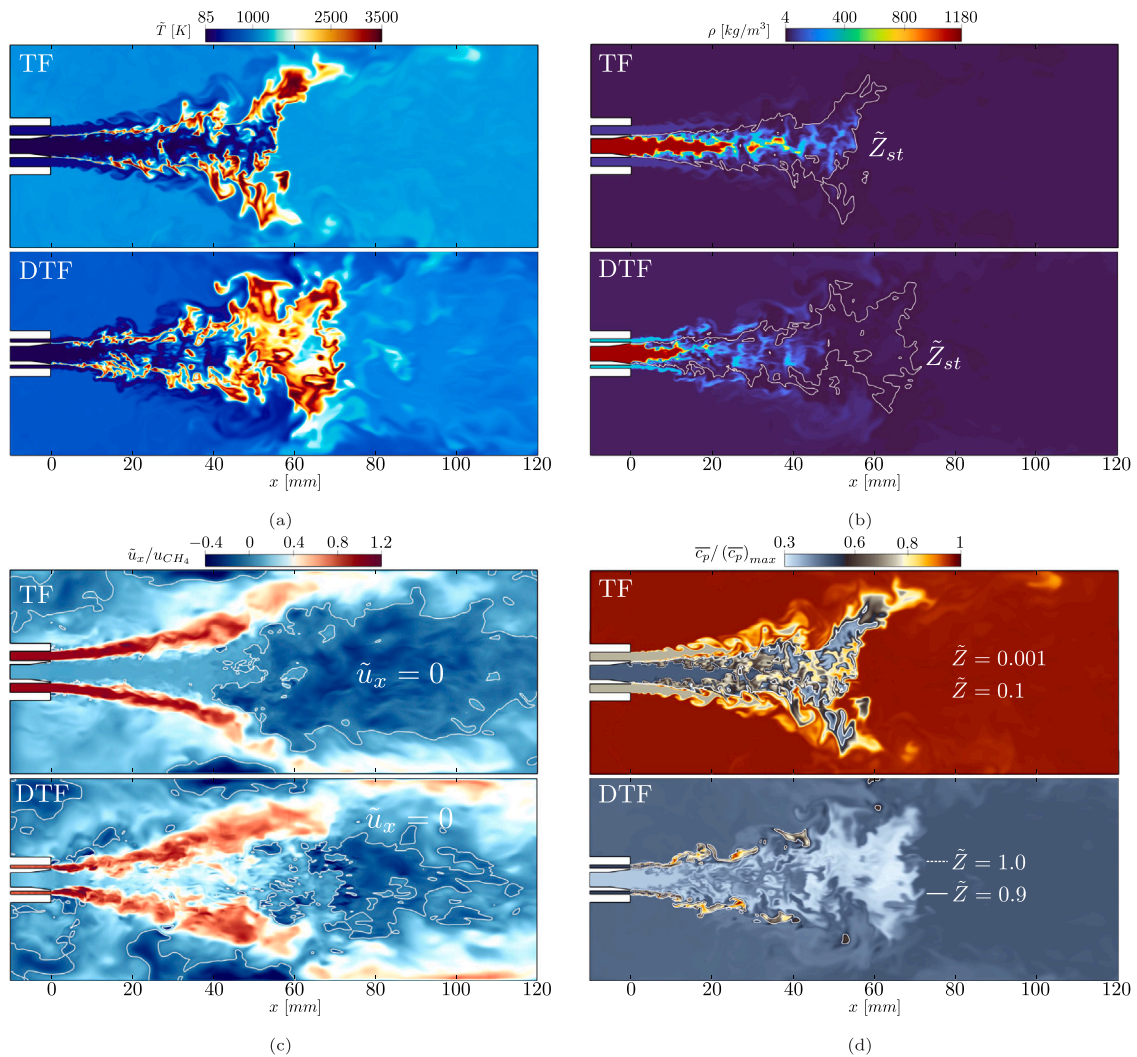


Fig. 10. Comparison between transcritical (top-panel) and doubly-transcritical (bottom-panel) flames with the SFM model. Longitudinal slices of instantaneous (a) temperature, (b) density, (c) axial velocity, and (d) isobaric-specific heat. The white solid isoline in (b) denotes the stoichiometric mixture fraction $\tilde{Z}_{st} = 0.2$ and in (b) $\tilde{u}_x = 0$ m/s. The white solid and dashed isolines in (d) enclosed the pseudo-boiling region.

recirculation zone is more compact and less extended towards the chamber wall, and the $\tilde{u}_x = 0$ m/s contour appears more fragmented and dispersed.

Finally, Fig. 10(d) compares the normalized isobaric specific heat $\bar{c}_p / (\bar{c}_p)_{max}$. From this figure, the analysis made in the previous section and shown in Fig. 9 can be placed in physical space. In particular, in the TF, the white solid isoline for $\tilde{Z} = 0.001$ and dashed for $\tilde{Z} = 0.1$, enclosed the entire pseudo-boiling region (see also the scale in Fig. 9). As previously discussed, the small-scale SGS mixing modeled by $\tilde{Z}''^2$ implies that in this region, the values of the normalized isobaric specific heat are between 0.6–0.8, while the maximum values are observed in the flame zone and where the flow is homogeneous, e.g., in the recirculation zone and downstream of the flame.

On the other hand, for the DTF, the pseudo-boiling region is enclosed between the white isoline of $\tilde{Z} = 1.0$ and $\tilde{Z} = 0.9$. An interesting observation can be made regarding the pseudo-boiling phenomena of cryogenic methane. The high values of $\bar{c}_p$ mark the coaxial injection, with the maximum values primarily occurring in the secondary shear layer between the cryogenic methane and the recirculating combustion products. In the next section, this unique aspect will be further discussed concerning the pseudo-boiling rate.

Fig. 11 compare the time-averaged flow fields of the investigated cases using both mixing models. The results of the averaged temperature field $\langle \tilde{T} \rangle$ for the TF shown in Fig. 11(a) indicate a similar overall

flame morphology between the mixing models, both predicting the same temperature where the flow is homogeneous, indicating that the overall mixing efficiency is not affected by the mixing rules. Also, since real fluid effects are not expected to occur within the gas-like methane, the elongation and expansion towards the wall of the coaxial jet do not appear to be altered. On the other hand, the main difference is observed in the cold core extension, which seems slightly longer with the SFM. As a consequence, the flame length, marked by the stoichiometric mixture fraction $\tilde{Z}_{st} = 0.2$ and defined as the distance from the injection plate, is higher, with a percentage deviation between the models of 13.7%. In addition, using MFM results in a flame pattern that is more confined to the central axis, especially in the final part of the flame. The higher radial spreading with SFM is also observed in the averaged axial velocity fields $\langle \tilde{u}_x \rangle$ in Fig. 11(b), the intense expansion of the oxidizer flow results in higher values of the axial velocity near the chamber wall.

Similar observations are reflected in the DTF, despite the similar flame topology, the flame length and spreading in the radial direction are restrained when using the MFM mixing model, which confirmed the overall flow compactness when using this model.

As reported in earlier works [55–57,69,81,87,90], we qualitatively compare in Fig. 12 the time-averaged numerical $\langle \text{OH}^* \rangle$ (top-panel) emission with the experimental [36] OH^* (bottom-panel) reconstructed radiation signal of the TF case. Since OH mass fraction is not repre-

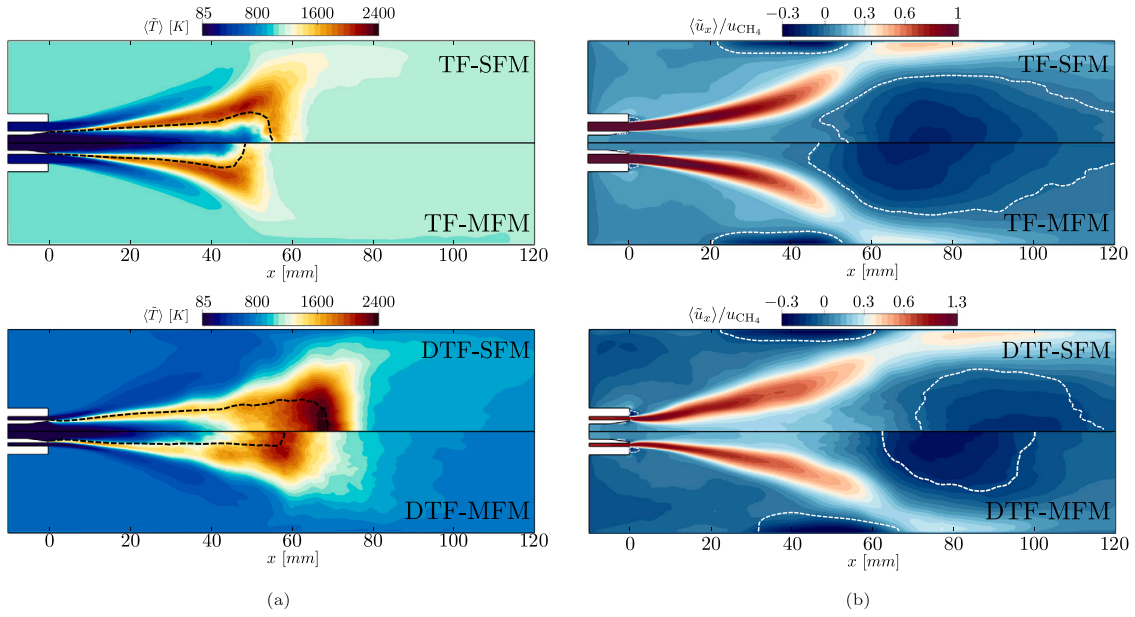


Fig. 11. Comparison between transcritical (top-panel) and doubly-transcritical (bottom-panel) flames with the SFM and MFM mixing model. Longitudinal slices of time-averaged (a) temperature and (b) axial velocity. The black dashed isoline in (a) denotes the stoichiometric mixture fraction $\langle \bar{Z}_{st} \rangle = 0.2$. The white dashed isoline in (b) denotes average null velocity $\langle \bar{u}_x \rangle = 0$ m/s.

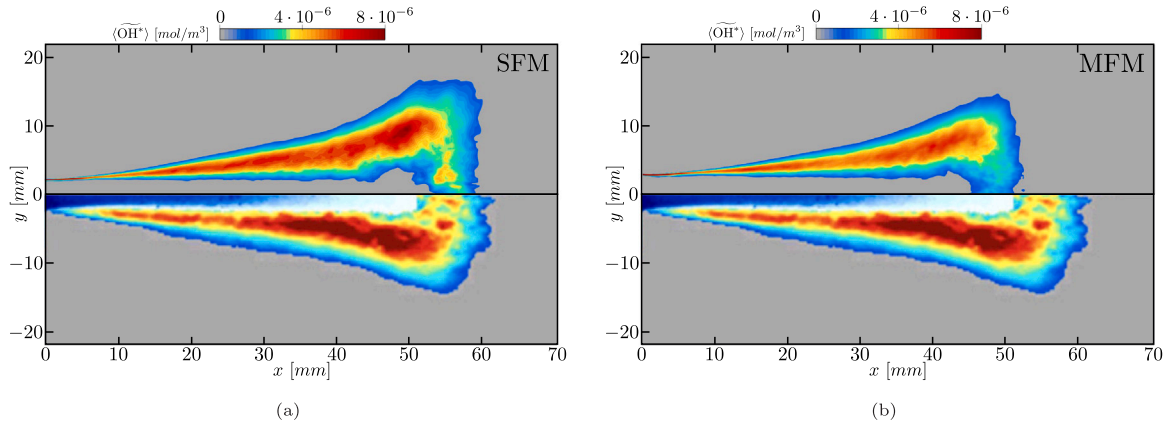


Fig. 12. Comparison with experimental visualizations of the TF case. Time-averaged numerical $\langle \overline{OH^*} \rangle$ concentration (upper half), and experimental [36] Abel-transformed OH^* emission (lower-half), for (a) SFM model, and (b) MFM model.

sentative for OH^* emissions, to ensure a precise comparison between experimental and numerical results, we employed the partial equilibrium method developed by Fiala and Sattelmayer [106] to compute the OH^* emission. The characteristic flame topology discussed for Fig. 11(a) is reflected with Abel transformed experimental OH^* emission. Both mixing models result in reasonable agreements with the experiments, and the general flame structure is properly retrieved. As observed for the temperature fields, in the near injection region, the flame is characterized for both models by a constant spreading angle and a slight underestimate of the flame thickness, while the main differences arise at the flame end, where the SFM is in good agreement concerning the flame length. Still, using the MFM leads to a more compact flame with a lower lateral spreading.

Note that this comparison cannot lead to a classification of the predictive performance of the mixing models w.r.t the experimental image since the precise dimensions of the experimental observation area are not disclosed, preventing a more detailed quantitative assessment. For both, the characteristic features and flame proportions are well reproduced by the LES, validating the solver and the methodology.

5.3. Statistical analysis of turbulent pseudo-boiling rate

In this section, we examine the statistical characteristics of the turbulent pseudo-boiling rate using LES data from the TF and DTF cases. Fig. 13 presents the area-weighted PDF of the mass flux per unit area across iso-surfaces selected following the methodology proposed in Section 5.1. Specifically, the first column shows the total J_{pb} , while the middle and right columns display its decomposition into the curvature J_{pb}^{CRV} and normal diffusion component J_{pb}^{ND} , as defined in Eq. (24).

For the LOx-side of the TF in Fig. 13(a), the PDFs of the total turbulent pseudo-boiling rate are asymmetric, showing a positive skewness for both mixing models. This indicates that, on average, the mass flux is directed from cold to hot regions due to flame heating, as expected. The left tails show minimal differences, with both distributions overlapping for negative values of J_{pb} . Such negative fluxes arise from the entrainment of light, warm gas pockets into the LOx core through unsteady mixing, as previously discussed in [17]. Differences between the mixing models emerge in the right tail: the MFM distribution decays more slowly beyond $J_{pb} = 1 \text{ kg/m}^2 \text{ s}$, indicating a higher likelihood of high positive pseudo-boiling events. Moreover, the curvature component

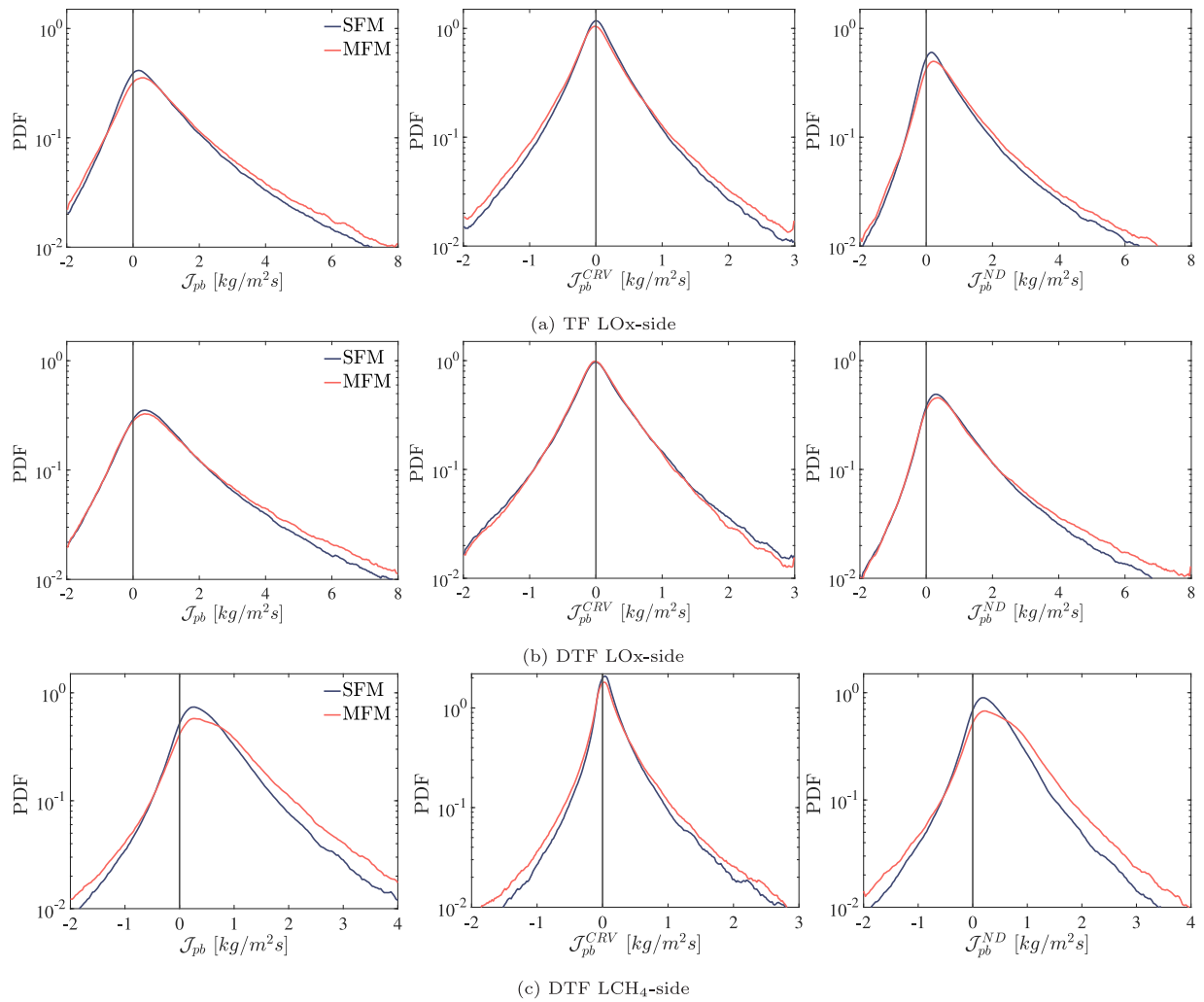


Fig. 13. Statistical features of the turbulent pseudo-boiling rate. Probability density function of J_{pb} (left-column), J_{pb}^{CRV} (middle-column), and J_{pb}^{ND} (right-column), using the SFM (blue-lines) and MFM (red-lines) models for: (a) TF LOx-side, (b) DTF LOx-side, and (c) DTF LCH₄-side. The selected pseudo-boiling iso-surfaces are those reported in Fig. 9.

appears more resistant to turbulent motions, yielding a narrower and nearly symmetric shape with a pronounced peak for near-zero pseudo-boiling events. The statistical analysis of the second term confirms that the dominant contribution to the total pseudo-boiling rate comes from the normal diffusion component J_{pb}^{ND} . This result aligns with previous experimental [34–36] and numerical [17,38] studies on cryogenic jets, suggesting that mass transfer under transcritical conditions is mainly governed by normal diffusion fluxes induced by the non-premixed flame between the two propellant streams. The PDFs for the oxidizer-rich side of the DTF in Fig. 13(b) show that pseudo-boiling occurs in a statistically similar way in both TF and DTF. This consistency is expected, as both cases share identical LOx injection thermodynamic conditions, with the only differing parameter being the velocity ratio due to the injector design.

The results for the LCH₄-side in Fig. 13(c) indicate slight differences in the pseudo-boiling mechanism of the cryogenic methane. The overall distribution trends and decomposition suggest the same physical interpretation discussed for the LOx-side. Nonetheless, differences arise in the pseudo-boiling rate values: the most likely value is lower w.r.t the oxidizer counterpart. This behavior can be explained by the heat shield phenomenon resulting from the inherently higher, w.r.t the oxidizer-rich side, isobaric-specific heat (see Fig. 9(b)). The mass transfer rate is moderately reduced due to the thermodynamic conditions. This region acts as a thermal barrier, limiting the heating of the cryogenic methane.

Overall, the results from the laminar and turbulent pseudo-boiling analysis in mixture fraction space (Figs. 8, 9), the time-averaged flow-fields (Figs. 11, 12), and the PDFs (Fig. 13) suggest that the rate at which both LOx-side and LCH₄-side pseudo-boils is affected by the mixing models, with the SFM being the one that leads to lower values.

To quantitatively assess these differences and clarify the physical mechanisms driving the pseudo-boiling rates—and potentially the observed flame pattern variations—Fig. 14 reports the mean pseudo-boiling rate $\langle J_{pb} \rangle$ and total pseudo-boiling area A_{pb} at different time instants for both mixing models. Each time t^* corresponds to a physical time of 2 ms. Using the decomposition in Eq. (26), the total rate is split into its laminar (resolved) component $\langle J_{pb}^{lam} \rangle$ and the SGS contribution $\langle J_{pb}^{sgs} \rangle$, hence the percentages given in the stacked bar denote their relative contributions.

Focusing on the LOx-side of the TF in Fig. 14(a), for all time instants considered, the total flux is appreciably higher with the MFM model than with the SFM, their ratio being between 1.08–1.4. As expected, the SGS contribution dominates in both models. However, the laminar component remains significant, accounting for 42%–46% of the total flux, supporting the argument that even in turbulent conditions, the real-fluid laminar diffusion coefficient – and thus variables such as c_p , λ that influence it – can have a non-negligible effect on the pseudo-boiling rates in well-resolved LES, as in the present work. Specifically, the MFM models yield a systematically higher resolved-scale contribution than the SFM over the entire time window. The physical mechanism

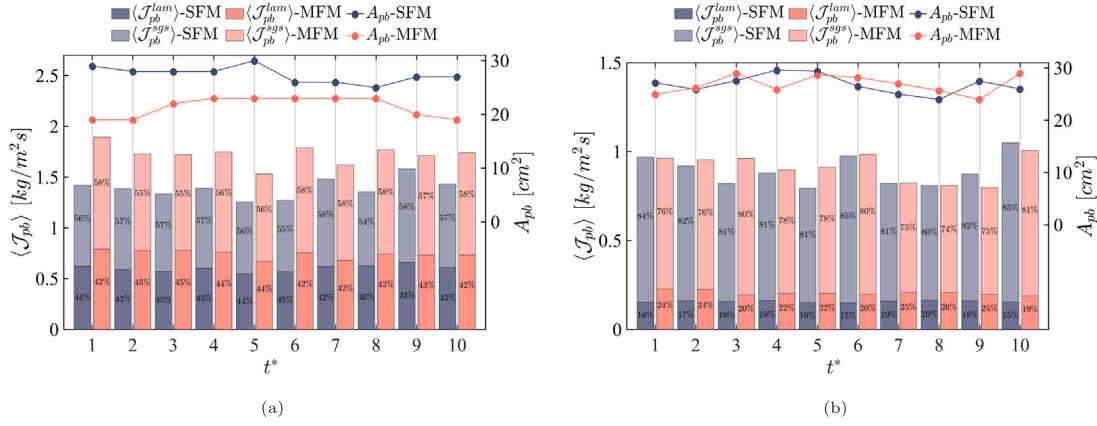


Fig. 14. Comparison of the mean resolved $\langle J_{pb}^{lam} \rangle$ and SGS $\langle J_{pb}^{sgs} \rangle$ pseudo-boiling rates and the total area A_{pb} using the SFM and MFM models for (a) TF LOx-side, (b) DTF LCH₄-side.

driving this difference can be attributed to a consistently larger resolved diffusivity $\bar{D}$ at the pseudo-boiling location, for the reason discussed in Section 5.1. The ratio of resolved diffusivities between the two models remains consistently above unity, with MFM values exceeding those of SFM by a factor of 1.05 to 1.11, contributing to the observed enhancement of the resolved pseudo-boiling flux term J_{pb}^{lam} .

On the other hand, the SGS contribution J_{pb}^{sgs} is consistently higher in the MFM than in the SFM. Again, this behavior is due to a model-induced effect resulting from the definition of the SGS diffusivity, $D^{sgs} = \bar{\rho} \nu^{sgs}$. While the SGS viscosity ν^{sgs} remains largely similar between the two models, the filtered density $\bar{\rho}$ is systematically higher in the MFM by up to about 20% on the selected pseudo-boiling surfaces. As a result, the total pseudo-boiling area A_{pb} is smaller with MFM. Assuming similar resolved surface wrinkling for both models, this result suggests that the SFM delays the jet's thermal breakup. Consequently, an extended LOx-core develops, which can explain the differences observed in Fig. 11. Moreover, as discussed in Section 5.1, MFM predicts a higher thermal diffusivity across the whole mixture fraction space, enhancing mass transfer and likely contributing to the moderate flame length reductions observed.

Focusing on the LCH₄-side of the DTF, the differences between SFM and MFM are smaller than those observed on the LOx-side. Although MFM yields a marginally higher mean total fluxes, it does not systematically dominate at all instants. As previously observed, MFM consistently leads to a larger resolved term than SFM for the reasons discussed so far, but in this case, it accounts only for about 16%–26% of the total flux. The lower values of $\langle J_{pb}^{lam} \rangle$ on this side reflect methane's higher thermal inertia, which inherently limits the pseudo-boiling rate under thermal gradients. However, a direct comparison with the oxidizer side is not straightforward, as the pseudo-boiling flux depends on local mixture fraction gradients, which in turn are shaped by the turbulent flowfield, the flame structure, and its interaction with turbulence. These factors introduce spatially varying scalar gradients that modulate J_{pb} beyond what can be inferred from the fluid properties alone. The SGS term shows comparable mean values to the LOx-side, but no systematic dominance emerges, with MFM/SFM ratios hovering around unity over the entire time window. Similarly, no clear trend is observed in A_{pb} , with neither model consistently yielding larger values over time.

In summary, for both sides, the mixing models have a clear impact on the resolved pseudo-boiling rate term due to the difference in local thermal diffusivity. In contrast to what was observed for the LOx side, the reduced model sensitivity on the SGS term of the LCH₄ side does not lead to a systematic trend of the pseudo-boiling mechanism that could be reflected in the total area that exhibits the pseudo-phase transition. Concerning this latter aspect, to improve its characterization, Fig. 15 shows the mean pseudo-boiling rate and area in terms of axial distance from the injector. Specifically, we subdivide the pseudo-boiling

interface into slabs of length equal to LOx diameter d_2 . Thus, each bar corresponds to the mean $\langle J_{pb} \rangle$ and the area of the interface portion enclosed within the slab. For the sake of figure readability only a time instant is considered, notably the last one of Fig. 14.

The axial evolution of A_{pb} for the LOx-side indicates that up to $x/d_2 = 8$ is slightly higher with the SFM, but the trend and the estimates between the models are closer. Since the slab length is fixed, this result suggests that the surface wrinkling induced by the models is similar and consequently reinforces the interpretation that the SFM leads to a delayed jet's thermal breakup, preserving a longer dense core. In fact, with the MFM model, from $x/d_2 = 9$, the area of the considered interface portion rapidly decreased up to $x/d_2 = 11$, where the effective pseudo-boiling surface no longer exists. Again, the MFM consistently predicts higher values of $\langle J_{pb} \rangle$ along the entire domain. These observations are systematic for all time instants.

In contrast, on the LCH₄-side, the axial evolution of A_{pb} shows no clear distinction between the models in the upstream region up to $x/d_2 = 6$. In this range, the extent of surface wrinkling appears to depend on the specific interface slab portion considered. Beyond $x/d_2 = 6$, both models predict a rapid decline in A_{pb} , with the SFM showing lower values. Nonetheless, the pseudo-boiling interface, and consequently the axial extension of the cryogenic methane, remains visible in both cases up to $x/d_2 = 9$. Similar to the geometrical trend, the evolution of $\langle J_{pb} \rangle$ does not exhibit a clear distinction of one mixing model over the other, and both depend on the specific time instant considered. This lack of systematic trend reflects the weaker sensitivity of the pseudo-boiling mechanism to the mixing model in this region.

To better understand the pseudo-boiling mechanism on the LCH₄-side, the following analysis focuses on the interplay between thermodynamic conditions and turbulence. From the PDFs in Fig. 13, it is observed that, unlike the LOx-side, which shows a linear decay on the right tail, the PDF of the total turbulent pseudo-boiling rate and normal diffusion component on the LCH₄-side exhibits a slope change, hinting an inherent asymmetry in the pseudo-boiling mechanism of cryogenic methane. To further investigate this aspect, the upper panel of Fig. 16 shows the $\tilde{Z}_{pb}$ iso-surfaces, color-coded by the resolved isobaric specific heat, while the lower panel presents the joint PDF (JPDF) of $(J_{pb}, \bar{c}_p)$.

For the LOx jet of the TF in Fig. 16(a), the JPDF exhibits unimodal behavior with the most probable events clustered around the effective turbulent pseudo-boiling specific heat $(\bar{c}_p)_{pb}$, and a gradual probability decay as $\bar{c}_p$ varies. In contrast, the coaxial LCH₄ jet of the DTF exhibits two distinct peaks. In particular, since LCH₄ is a coaxial jet, it pseudo-boils across a surface with two sides, an inner and an outer side. This bimodal behavior reflects the different thermodynamic conditions in the inner (ISL) and outer (OSL) shear layers. Specifically, the ISL, located between the established flame and the methane jet, is characterized by low and uniform values of $\bar{c}_p$ comprised in a range of

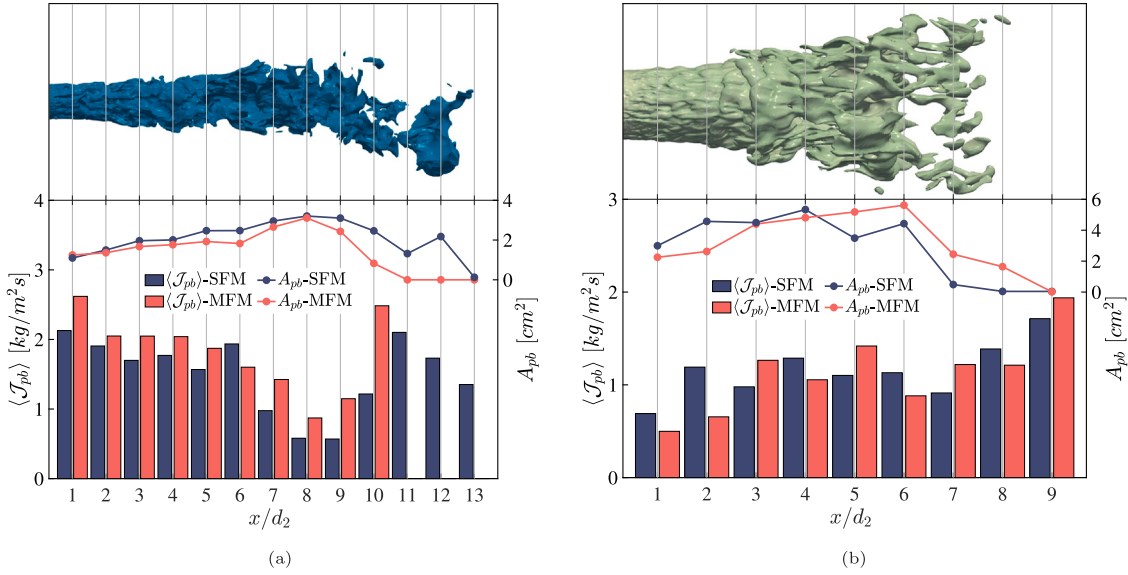


Fig. 15. Comparison of the mean $\langle J_{pb} \rangle$ pseudo-boiling rate and area A_{pb} using the SFM and MFM models as a function of the axial distance from the injector for (a) TF LOx-side, (b) DTF LCH₄-side.

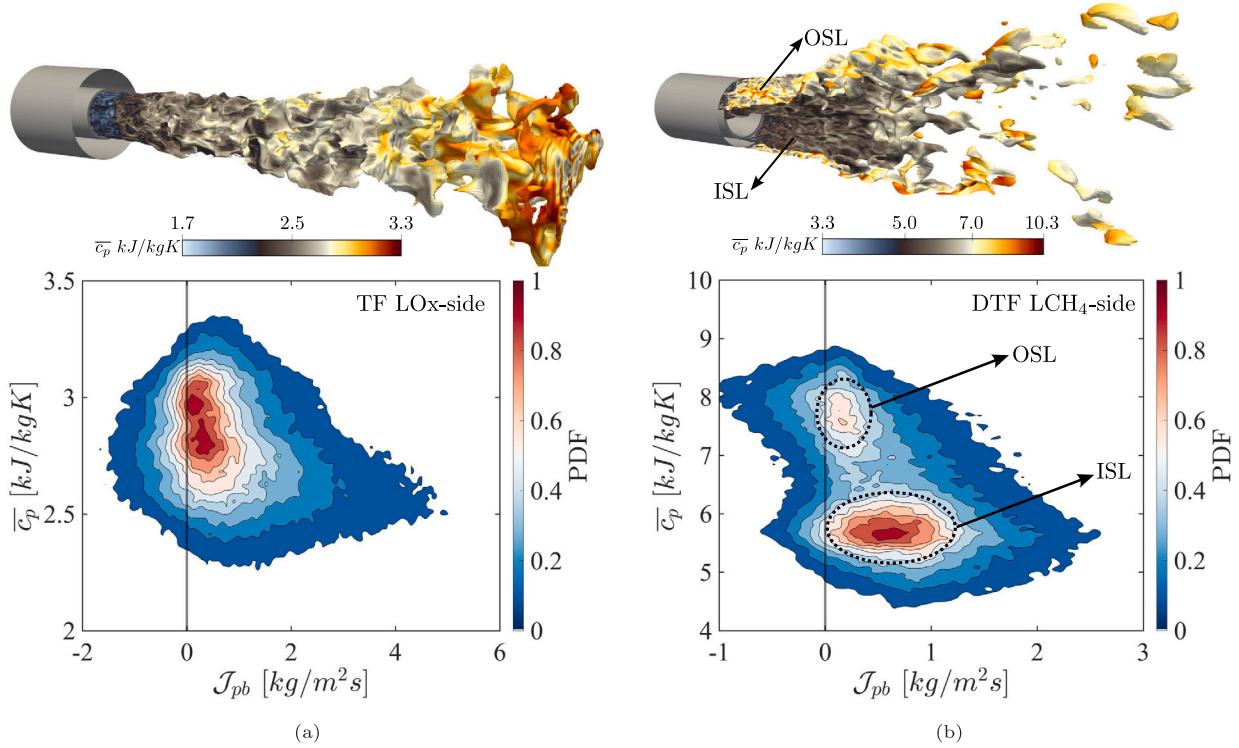


Fig. 16. Statistical features of the turbulent pseudo-boiling rate. Joint probability density function between J_{pb} and $\bar{c}_p$, for (a) TF LOx-side and (b) DTF LCH₄-side. The upper panel shows the respective $\tilde{Z}_{pb}$ iso-surfaces color-coded with the isobaric specific heat.

$\approx 5\text{--}6.5$ kJ/kg K. This leads to a broad region of statistically significant high mass flux events. Conversely, the OSL, located between the outer methane jet and the recirculation zone, has higher $\bar{c}_p$ values, resulting in a second, narrower cluster associated with lower mass flux.

This distinctive feature, arising from the interplay between turbulence and thermodynamic conditions, is illustrated in Fig. 17, which reports, for the DTF, longitudinal slices of the instantaneous resolved scalar dissipation rate $\tilde{\chi} = 2(\bar{D} + D^{sgs})(\nabla \tilde{Z})^2$ (top panel of Fig. 17(a)) and the normalized mixture fraction variance $\tilde{Z}_n''^2$ (bottom panel). The white isoline denotes the $\tilde{Z}_{pb}$ iso-surface. As can be observed, the ISL

is marked by high values of $\tilde{\chi}$, reflecting intense local strain induced by rapid thermal expansion due to the presence of the flame. This strong turbulence–chemistry interaction in elevated $\tilde{Z}_n''^2$ values and a consequent smearing of $\bar{c}_p$ (see Fig. 9). In contrast, the OSL features $\tilde{\chi}$ values nearly two orders of magnitude lower, leading to reduced $\tilde{Z}_n''^2$ and a higher thermal inertia of the interface.

This transport process at the bounding interface between the turbulent and non-turbulent flow regions reflects the inherent asymmetry between the two pseudo-boiling mechanisms, where the internal jet, both for LOx and LCH₄ sides, is strongly influenced by shear-layer

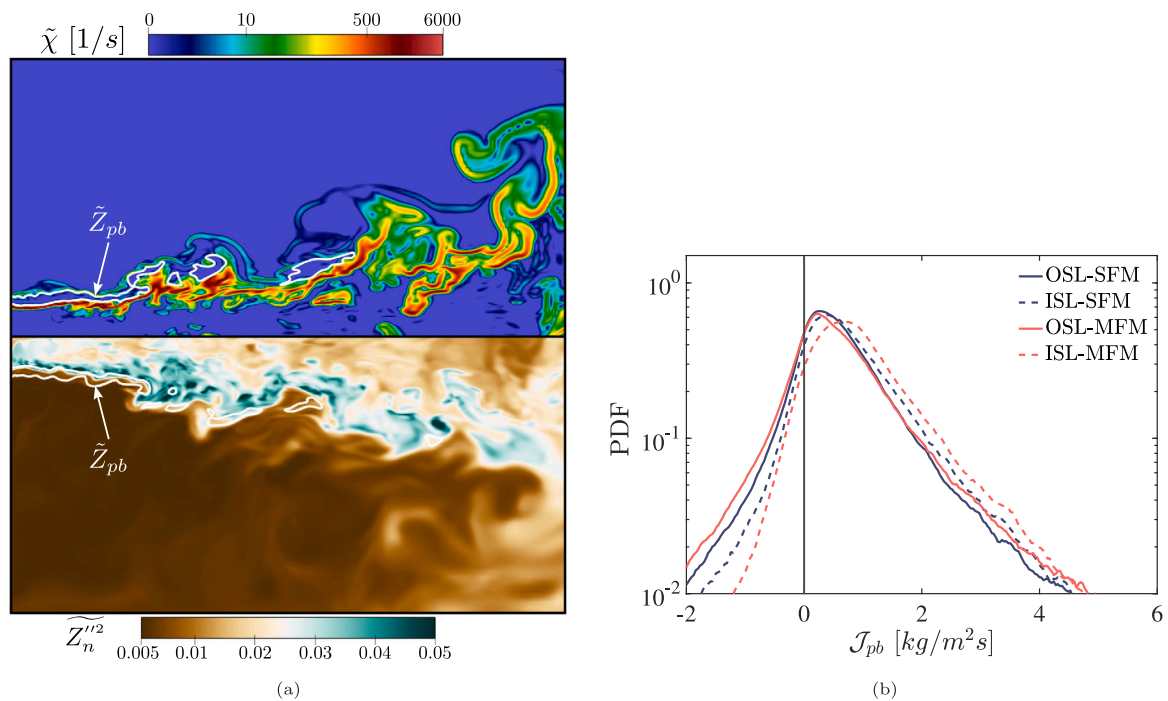


Fig. 17. Interplay between turbulent flow field and pseudo-boiling mechanism for DTF LCH₄-side. (a) Comparison between instantaneous scalar dissipation rate field $\tilde{\chi}$ (top half) and normalized mixture fraction variance $\tilde{Z}_n''^2$ (bottom half). The white solid isoline denotes the $\tilde{Z}_{pb}$ interface. (b) Probability density function of J_{pb} for the OSL (solid-lines) and ISL (dashed-lines) using the SFM (blue-lines) and MFM (red-lines) models.

dynamics. In contrast, the coaxial outer LCH₄ side shows restrained aerodynamic strain levels. Circumscribing the pseudo-boiling rate contribution between the ISL and OSL of the DTF case, the PDFs distribution displayed in Fig. 17(b) indicates that for both mixing models, the ISL distributions are moderately wider compared to the OSL one. This observation confirms that significant straining (elevated values of $\tilde{\chi}$) of the interface at pseudo-boiling conditions is needed to counteract its insulating effects.

6. Conclusions

The turbulent pseudo-boiling of cryogenic propellants has been investigated in transcritical and doubly-transcritical LOx–methane flames under technologically relevant conditions for liquid rocket engines.

A flamelet-based approach, coupled with different mixing rules for estimating thermodynamic properties and high-resolved large-eddy simulations, was employed to gain insight into the different flame behaviors. The characteristic flame morphologies at transcritical and doubly-transcritical conditions have been discussed, highlighting the qualitative and quantitative differences when using different mixing models. Given the limited number of studies addressing scenarios where both propellants are injected at cryogenic temperatures, the present results are expected to provide valuable insights to the community.

A consistent definition of the turbulent pseudo-boiling rate has been introduced for the combustion-LES framework. Since the actual pseudo-boiling typically occurs at the sub-grid scale level, a strategy to identify the effective pseudo-boiling state at the LES grid level has been developed and discussed. Using this framework, with laminar flamelet calculations and the developed LES datasets, it is found that the use of different mixing rules affects the thermal breakup of the jet. This occurs even though the pseudo-phase transition from liquid-like to gas-like conditions of the propellant streams occurs almost as a pure fluid.

We have carried out a thorough statistical analysis of the turbulent pseudo-boiling rate. Its decomposition into normal and curvature components indicates that mass transfer under transcritical conditions

is primarily driven by normal diffusion fluxes. This result confirms that the cryogenic combustion process and flame morphologies are diffusion-limited, as they are governed by mass fluxes from the dense regions to the lighter gas streams. The analysis has also revealed that the two flames pseudo-boil the cryogenic oxidizer-rich mixture at similar rates. In contrast, the pseudo-boiling rate on the fuel-rich side in the doubly-transcritical case is significantly lower w.r.t the oxygen side. This behavior is ascribed to the higher constant pressure specific heat of the methane and the different interaction with shear-generated turbulence. A detailed quantification of the resolved and SGS contributions to the turbulent pseudo-boiling rate has been performed. For the cryogenic oxidizer-rich mixture, the resolved component plays a non-negligible role and is consistently higher when using the MFM due to the enhanced thermal diffusivity. On the other hand, the SGS contribution is mainly influenced by local density variations, which are also affected by the choice of mixing model. On the LCH₄ side, a similar model sensitivity is observed despite the smaller magnitude of the resolved contribution; however, no systematic dominance for the SGS flux emerges.

Additional analysis of the multivariate statistics between thermal inertia and relative mass fluxes has highlighted the difference in mixing efficiency between the inner and outer coaxial shear layers. As a result, to efficiently pseudo-boil the propellants in cryogenic injection and promote the establishment of a compact flame, significant strain is required at the effective site of pseudo-vaporization to counteract its insulating effect.

CRediT authorship contribution statement

D. Cavalieri: Writing – original draft, Software, Methodology, Investigation, Conceptualization. **D. Schintu:** Writing – review & editing, Software, Methodology. **P.P. Ciottoli:** Writing – review & editing, Supervision. **P.E. Lapenna:** Writing – review & editing, Supervision, Conceptualization. **F. Creta:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] M. Pizzarelli, F. Battista, Oxygen-methane rocket thrust chambers: Review of heat transfer experimental studies, *Acta Astronaut.* 209 (2023) 48–66, <http://dx.doi.org/10.1016/j.actaastro.2023.04.028>.
- [2] R. Roback, E.J. Szelata, L.J. Spadaccini, *Deposit Formation in Hydrocarbon Rocket Fuels*, NASA Contractor Report 165405, NASA Lewis Research Center, Cleveland, OH, 1981.
- [3] P. Caisso, A. Souchier, C. Rothmund, P. Alliot, C. Bonhomme, W. Zinner, R. Parsley, T. Neill, S. Forde, R. Starke, W. Wang, M. Takahashi, M. Atsumi, D. Valentian, A liquid propulsion panorama, *Acta Astronaut.* 65 (11) (2009) 1723–1737, <http://dx.doi.org/10.1016/j.actaastro.2009.04.020>.
- [4] W. Mayer, H. Tamura, Propellant injection in a liquid oxygen/gaseous hydrogen rocket engine, *J. Propuls. Power* 12 (1996) 1137–1147, <http://dx.doi.org/10.2514/3.24154>.
- [5] W.O.H. Mayer, A.H.A. Schik, B. Vielle, C. Chauveau, I. Gokalp, D.G. Talley, R.D. Woodward, Atomization and breakup of cryogenic propellants under high-pressure subcritical and supercritical conditions, *J. Propuls. Power* 14 (5) (1998) 835–842, <http://dx.doi.org/10.2514/2.5348>.
- [6] R. Branam, W. Mayer, Characterization of cryogenic injection at supercritical pressure, *J. Propuls. Power* 19 (3) (2003) 342–355, <http://dx.doi.org/10.2514/2.6138>.
- [7] M. Oschwald, J.J. Smith, R. Branam, J. Hussong, A. Schik, B. Chehroudi, D. Talley, Injection of fluids into supercritical environments, *Combust. Sci. Technol.* 178 (1–3) (2006) 49–100, <http://dx.doi.org/10.1080/00102200500292464>.
- [8] S. Candel, M. Juniper, G. Singla, P. Scoufflaire, C. Rolon, Structure and dynamics of cryogenic flames at supercritical pressure, *Combust. Sci. Technol.* 178 (1–3) (2006) 161–192, <http://dx.doi.org/10.1080/00102200500292530>.
- [9] J.C. Oefelein, V. Yang, Modeling high-pressure mixing and combustion processes in liquid rocket engines, *J. Propuls. Power* 14 (5) (1998) 843–857, <http://dx.doi.org/10.2514/2.5349>.
- [10] H. Meng, V. Yang, A unified treatment of general fluid thermodynamics and its application to a preconditioning scheme, *J. Comput. Phys.* 189 (1) (2003) 277–304, [http://dx.doi.org/10.1016/S0021-9991\(03\)00211-0](http://dx.doi.org/10.1016/S0021-9991(03)00211-0).
- [11] D.Y. Peng, D.B. Robinson, A new two-constant equation of state, *Ind. Eng. Chem. Res.* 15 (1) (1976) 59–64, <http://dx.doi.org/10.1021/i160057a011>.
- [12] G. Soave, Equilibrium constants from a modified Redlich-Kwong equation of state, *Chem. Eng. Sci.* 27 (6) (1972) 1197–1203, [http://dx.doi.org/10.1016/0009-2509\(72\)80096-4](http://dx.doi.org/10.1016/0009-2509(72)80096-4).
- [13] B.E. Poling, J.M. Prausnitz, J.P. O'Connell, *The Properties of Gases and Liquids*, fifth ed., McGraw-Hill Education, 2001.
- [14] S.K. Kim, H.S. Choi, Y. Kim, Thermodynamic modeling based on a generalized cubic equation of state for kerosene/LOx rocket combustion, *Combust. Flame* 159 (3) (2012) 1351–1365, <http://dx.doi.org/10.1016/j.combustflame.2011.10.008>.
- [15] L. Jofre, J. Urzay, Transcritical diffuse-interface hydrodynamics of propellants in high-pressure combustors of chemical propulsion systems, *Prog. Energy Combust. Sci.* 82 (2021) 100877, <http://dx.doi.org/10.1016/j.pecs.2020.100877>.
- [16] D.T. Banuti, Crossing the Widom-line – Supercritical pseudo-boiling, *J. Supercrit. Fluids* 98 (2015) 12–16, <http://dx.doi.org/10.1016/j.supflu.2014.12.019>.
- [17] P.E. Lapenna, Characterization of pseudo-boiling in a transcritical nitrogen jet, *Phys. Fluids* 30 (7) (2018) 077106, <http://dx.doi.org/10.1063/1.5038674>.
- [18] F. Gorelli, M. Santoro, T. Scopigno, M. Krisch, G. Ruocco, Liquidlike behavior of supercritical fluids, *Phys. Rev. Lett.* 97 (2006) 245702, <http://dx.doi.org/10.1103/PhysRevLett.97.245702>.
- [19] T. Sato, M. Sugiyama, K. Itoh, K. Mori, T. Fukunaga, M. Misawa, T. Otomo, S. Takata, Structural difference between liquidlike and gaslike phases in supercritical fluid, *Phys. Rev. E* 78 (2008) 051503, <http://dx.doi.org/10.1103/PhysRevE.78.051503>.
- [20] G.G. Simeoni, T. Bryk, F.A. Gorelli, M. Krisch, G. Ruocco, M. Santoro, T. Scopigno, The Widom line as the crossover between liquid-like and gas-like behaviour in supercritical fluids, *Nat. Phys.* 6 (2010) 503–507, <http://dx.doi.org/10.1038/nphys1683>.
- [21] M. Raju, D.T. Banuti, P.C. Ma, M. Ihme, Widom lines in binary mixtures of supercritical fluids, *Sci. Rep.* 7 (2017) 3027, <http://dx.doi.org/10.1038/s41598-017-03334-3>.
- [22] D.T. Banuti, M. Raju, M. Ihme, Between supercritical liquids and gases – Reconciling dynamic and thermodynamic state transitions, *J. Supercrit. Fluids* 165 (2020) 104895, <http://dx.doi.org/10.1016/j.supflu.2020.104895>.
- [23] Y.J. Yul, The turbulent flows of supercritical fluids with heat transfer, *Annu. Rev. Fluid Mech.* (ISSN: 1545-4479) 45 (2013) 495–525, <http://dx.doi.org/10.1146/annurev-fluid-120710-101234>.
- [24] A. Urbano, F. Nasuti, Conditions for the occurrence of heat transfer deterioration in light hydrocarbons flows, *Int. J. Heat Mass Transfer* (ISSN: 0017-9310) 65 (2013) 599–609, <http://dx.doi.org/10.1016/j.ijheatmasstransfer.2013.06.038>.
- [25] M. Pizzarelli, F. Nasuti, M. Onofri, P. Roncioni, R. Votta, F. Battista, Heat transfer modeling for supercritical methane flowing in rocket engine cooling channels, *Appl. Therm. Eng.* (ISSN: 1359-4311) 75 (2015) 600–607, <http://dx.doi.org/10.1016/j.applthermaleng.2014.10.008>.
- [26] G. Dang, F. Zhong, Y. Zhang, X. Zhang, Numerical study of heat transfer deterioration of turbulent supercritical kerosene flow in heated circular tube, *Int. J. Heat Mass Transfer* (ISSN: 0017-9310) 85 (2015) 1003–1011, <http://dx.doi.org/10.1016/j.ijheatmasstransfer.2015.02.052>.
- [27] M. Pizzarelli, The status of the research on the heat transfer deterioration in supercritical fluids: A review, *Int. Comm. Heat Mass Transf.* (ISSN: 0735-1933) 95 (2018) 132–138, <http://dx.doi.org/10.1016/j.icheatmasstransfer.2018.04.006>.
- [28] F. Sun, Y. Li, O. Manca, G. Xie, On assessment of heat transfer deterioration of a channel with supercritical n-decane for scramjet engines cooling, *Int. J. Heat Mass Transfer* (ISSN: 0017-9310) 135 (2019) 782–795, <http://dx.doi.org/10.1016/j.ijheatmasstransfer.2019.02.027>.
- [29] F. Nasuti, M. Pizzarelli, Pseudo-boiling and heat transfer deterioration while heating supercritical liquid rocket engine propellants, *J. Supercrit. Fluids* (ISSN: 0896-8446) 168 (2021) 105066, <http://dx.doi.org/10.1016/j.supflu.2020.105066>.
- [30] N. Longmire, D.T. Banuti, Onset of heat transfer deterioration caused by pseudo-boiling in CO₂ laminar boundary layers, *Int. J. Heat Mass Transfer* (ISSN: 0017-9310) 193 (2022) 122957, <http://dx.doi.org/10.1016/j.ijheatmasstransfer.2022.122957>.
- [31] R. Xiao, L. Chen, Z. Zhang, H. Lv, Y. Hou, Study on the heat transfer deterioration of supercritical nitrogen in a vertical tube using a pseudo-boiling model, *Int. J. Heat Mass Transfer* (ISSN: 0017-9310) 238 (2025) 126416, <http://dx.doi.org/10.1016/j.ijheatmasstransfer.2024.126416>.
- [32] M. Oschwald, A. Schik, Supercritical nitrogen free jet investigated by spontaneous Raman scattering, *Exp. Fluids* 27 (1999) 497–506, <http://dx.doi.org/10.1007/s003480050374>.
- [33] W. Mayer, J. Telaar, R. Branam, G. Schneider, J. Hussong, Raman measurements of cryogenic injection at supercritical pressure, *Heat Mass Transf.* 39 (2003) 709–719, <http://dx.doi.org/10.1007/s00231-002-0315-x>.
- [34] M. Juniper, A. Tripathi, P. Scoufflaire, J.C. Rolon, S. Candel, Structure of cryogenic flames at elevated pressures, *Proc. Combust. Inst.* 28 (1) (2000) 1103–1109, [http://dx.doi.org/10.1016/S0082-0784\(00\)80320-3](http://dx.doi.org/10.1016/S0082-0784(00)80320-3).
- [35] M. Habiballah, M. Orain, F. Grisch, L. Vingert, P. Gicquel, Experimental studies of high-pressure cryogenic flames on the Mascotte facility, *Combust. Sci. Technol.* 178 (1–3) (2006) 101–128, <http://dx.doi.org/10.1080/00102200500294486>.
- [36] G. Singla, P. Scoufflaire, C. Rolon, S. Candel, Transcritical oxygen/transcritical or supercritical methane combustion, *Proc. Combust. Inst.* 30 (2) (2005) 2921–2928, <http://dx.doi.org/10.1016/j.proci.2004.08.063>.
- [37] G. Ribert, N. Zong, V. Yang, L. Pons, N. Darabiha, S. Candel, Counterflow diffusion flames of general fluids: Oxygen/hydrogen mixtures, *Combust. Flame* 154 (3) (2008) 319–330, <http://dx.doi.org/10.1016/j.combustflame.2008.04.023>.
- [38] L. Pons, N. Darabiha, G. Ribert S. Candel, V. Yang, Mass transfer and combustion in transcritical non-premixed counterflows, *Combust. Theory Model.* 13 (1) (2009) 57–81, <http://dx.doi.org/10.1080/13647830802368821>.
- [39] G. Lacaze, J.C. Oefelein, A non-premixed combustion model based on flame structure analysis at supercritical pressures, *Combust. Flame* 159 (6) (2012) 2087–2103, <http://dx.doi.org/10.1016/j.combustflame.2012.02.003>.
- [40] D.T. Banuti, P.C. Ma, J.P. Hickey, M. Ihme, Thermodynamic structure of supercritical LOX–GH2 diffusion flames, *Combust. Flame* 196 (2018) 364–376, <http://dx.doi.org/10.1016/j.combustflame.2018.06.016>.

- [41] P.E. Lapenna, P.P. Ciottoli, F. Creta, The effect of fuel composition on the non-premixed flame structure of LNG/LOx mixtures at supercritical pressure, in: 54th AIAA Aerospace Sciences Meeting, 54th AIAA Aerospace Sciences Meeting, 2016, pp. 1–11, <http://dx.doi.org/10.2514/6.2016-0690>.
- [42] P.E. Lapenna, P.P. Ciottoli, F. Creta, Unsteady non-premixed methane/oxygen flame structures at supercritical pressures, *Combust. Sci. Technol.* 189 (12) (2017) 2056–2082, <http://dx.doi.org/10.1080/00102202.2017.1358710>.
- [43] H. Tani, S. Teramoto, N. Yamanishi, K. Okamoto, A numerical study on a temporal mixing layer under transcritical conditions, *Comput. & Fluids* 85 (2013) 93–104, <http://dx.doi.org/10.1016/j.compfluid.2012.10.022>.
- [44] H. Terashima, M. Koshi, Unique characteristics of cryogenic nitrogen jets under supercritical pressures, *J. Propuls. Power* 29 (6) (2013) 1328–1336, <http://dx.doi.org/10.2514/1.B34942>.
- [45] F. Battista, F. Picano, C.M. Casciola, Turbulent mixing of a slightly supercritical van der Waals fluid at low-mach number, *Phys. Fluids* 26 (5) (2014) 055101, <http://dx.doi.org/10.1063/1.4873200>.
- [46] P.E. Lapenna, F. Creta, Mixing under transcritical conditions: An a-priori study using direct numerical simulation, *J. Supercrit. Fluids* 128 (2017) 263–278, <http://dx.doi.org/10.1016/j.supflu.2017.05.005>.
- [47] P.E. Lapenna, F. Creta, Direct numerical simulation of transcritical jets at moderate Reynolds number, *AIAA J.* 57 (6) (2019) 2254–2263, <http://dx.doi.org/10.2514/1.J058360>.
- [48] P.E. Lapenna, Thermodynamic small scales in transcritical turbulent jets, *AIAA J.* 59 (6) (2021) 2328–2332, <http://dx.doi.org/10.2514/1.J059664>.
- [49] C. Laurent, L. Esclapez, D. Maestro, G. Staffelbach, B. Cuenot, L. Selle, T. Schmitt, F. Duchaine, T. Poinso, Flame-wall interaction effects on the flame root stabilization mechanisms of a doubly-transcritical LO₂/LCH₄ cryogenic flame, *Proc. Combust. Inst.* 37 (4) (2019) 5147–5154, <http://dx.doi.org/10.1016/j.proci.2018.05.105>.
- [50] F. Monnier, G. Ribert, L.D. Duvilla, CH₄/O₂ supercritical flame structure and simulation, *Proc. Combust. Inst.* 40 (1) (2024) 105514, <http://dx.doi.org/10.1016/j.proci.2024.105514>.
- [51] F. Monnier, G. Ribert, Numerical simulations of supercritical CH₄/O₂ flame propagation in inhomogeneous mixtures following ignition, *Proc. Combust. Inst.* 39 (2) (2023) 2747–2755, <http://dx.doi.org/10.1016/j.proci.2022.07.213>.
- [52] T. Schmitt, Y. Méry, M. Boileau, S. Candel, Large-Eddy simulation of oxygen/methane flames under transcritical conditions, *Proc. Combust. Inst.* 33 (1) (2011) 1383–1390, <http://dx.doi.org/10.1016/j.proci.2010.07.036>.
- [53] N. Guézennec, M. Masquelet, S. Menon, Large eddy simulation of flame-turbulence interactions in a LOX-CH₄ shear coaxial injector, in: 50th AIAA Aerospace Sciences Meeting Including the New Horizons Forum and Aerospace Exposition, 50th AIAA Aerospace Sciences Meeting, 2012, pp. 1–17, <http://dx.doi.org/10.2514/6.2012-1267>.
- [54] L. Hakim, A. Ruiz, T. Schmitt, M. Boileau, G. Staffelbach, S. Ducruix, B. Cuenot, S. Candel, Large eddy simulations of multiple transcritical coaxial flames submitted to a high-frequency transverse acoustic modulation, *Proc. Combust. Inst.* 35 (2) (2015) 1461–1468, <http://dx.doi.org/10.1016/j.proci.2014.05.142>.
- [55] J. Zips, H. Müller, M. Pfitzner, Efficient thermo-chemistry tabulation for non-premixed combustion at high-pressure conditions, *Flow. Turbul. Combust.* 101 (2018) 821–850, <http://dx.doi.org/10.1007/s10494-018-9932-z>.
- [56] C. Traxinger, J. Zips, M. Pfitzner, Single-phase instability in non-premixed flames under liquid rocket engine relevant conditions, *J. Propuls. Power* 35 (4) (2019) 675–689, <http://dx.doi.org/10.2514/1.B37395>.
- [57] T. Schmitt, Large-eddy simulations of the mascot test cases operating at supercritical pressure, *Flow Turbul. Combust.* 105 (1) (2020) 159–189, <http://dx.doi.org/10.1007/s10494-019-00096-y>.
- [58] G. Lacaze, T. Schmitt, A. Ruiz, J.C. Oefelein, Comparison of energy-, pressure- and enthalpy-based approaches for modeling supercritical flows, *Comput. & Fluids* 181 (2019) 35–56, <http://dx.doi.org/10.1016/j.compfluid.2019.01.002>.
- [59] P.C. Ma, H. Wu, D.T. Banuti, M. Ihme, On the numerical behavior of diffuse-interface methods for transcritical real-fluids simulations, *Int. J. Multiph. Flow* 113 (2019) 231–249, <http://dx.doi.org/10.1016/j.ijmultiphaseflow.2019.01.015>.
- [60] C. Laurent, G. Staffelbach, F. Nicoud, T. Poinso, Heat-release dynamics in a doubly-transcritical LO₂/LCH₄ cryogenic coaxial jet flame subjected to fuel inflow acoustic modulation, *Proc. Combust. Inst.* 38 (4) (2021) 6375–6383, <http://dx.doi.org/10.1016/j.proci.2020.05.051>.
- [61] W.A. Cañas-Marín, U.E. Guerrero-Aconcha, J.D. Ortiz-Arango, Comparison of different cubic equations of state and combination rules for predicting residual chemical potential of binary and ternary Lennard–Jones mixtures: Solid-supercritical fluid phase equilibria, *Fluid Phase Equilib.* 234 (1) (2005) 42–50, <http://dx.doi.org/10.1016/j.fluid.2005.05.014>.
- [62] V. Giovangigli, L. Matuszewski, F. Dupoirieux, Detailed modeling of planar transcritical H₂–O₂–N₂ flames, *Combust. Theory Model.* 15 (2) (2011) 141–182, <http://dx.doi.org/10.1080/13647830.2010.527016>.
- [63] P.J. Linstrom, W.G. Mallard, NIST Chemistry WebBook, Technical Report NIST Standard Reference Database Number 69, National Institute of Standards and Technology, Gaithersburg, MD, 2024, <http://dx.doi.org/10.18434/T4D303>.
- [64] A. Burcat, B. Ruscic, Third Millennium Ideal Gas and Condensed Phase Thermochemical Database for Combustion (With Update from Active Thermochemical Tables), Technical Report, Argonne National Lab. (ANL), Argonne, IL (United States), 2005, <http://dx.doi.org/10.2172/925269>, URL <https://www.osti.gov/biblio/925269>.
- [65] T.H. Chung, M. Ajlan, L.L. Lee, K.E. Starling, Generalized multiparameter correlation for nonpolar and polar fluid transport properties, *Ind. Eng. Chem. Res.* 27 (4) (1988) 671–679, <http://dx.doi.org/10.1021/ie00076a024>.
- [66] J.F. Ely, H.J.M. Hanley, Prediction of transport properties. 1. Viscosity of fluids and mixtures, *Ind. Eng. Chem. Fund.* 20 (4) (1981) 323–332, <http://dx.doi.org/10.1021/i100004a004>.
- [67] D.T. Banuti, V. Hannemann, K. Hannemann, B. Weigand, An efficient multi-fluid-mixing model for real gas reacting flows in liquid propellant rocket engines, *Combust. Flame* 168 (2016) 98–112, <http://dx.doi.org/10.1016/j.combustflame.2016.03.029>.
- [68] J.C. Oefelein, Mixing and combustion of cryogenic oxygen-hydrogen shear-coaxial jet flames at supercritical pressure, *Combust. Sci. Technol.* 178 (1–3) (2006) 229–252, <http://dx.doi.org/10.1080/00102200500325322>.
- [69] T. Kim, Y. Kim, S.K. Kim, Effects of pressure and inlet temperature on coaxial gaseous methane/liquid oxygen turbulent jet flame under transcritical conditions, *J. Supercrit. Fluids* 81 (2013) 164–174, <http://dx.doi.org/10.1016/j.supflu.2013.05.011>.
- [70] R.B. Bird, W.E. Stewart, E.N. Lightfoot, Transport Phenomena, second ed., John Wiley & Sons Ltd., New York, 2002, <http://dx.doi.org/10.1115/1.1424298>.
- [71] R.J. Kee, J. Warnatz, J.A. Miller, Fortran Computer-Code Package for the Evaluation of Gas-Phase Viscosities, Conductivities, and Diffusion Coefficients, Technical Report, Sandia National Labs., Livermore, CA (USA), 1983, URL <https://www.osti.gov/biblio/6381554>.
- [72] N. Peters, Laminar flamelet concepts in turbulent combustion, Twenty-First Symp. (Int.) Combust. 21 (1) (1988) 1231–1250, [http://dx.doi.org/10.1016/S0082-0784\(88\)80355-2](http://dx.doi.org/10.1016/S0082-0784(88)80355-2).
- [73] A. Cuoci, A. Frassoldati, T. Faravelli, E. Ranzi, OpenSMOKE++: An object-oriented framework for the numerical modeling of reactive systems with detailed kinetic mechanisms, *Comput. Phys. Comm.* 192 (2015) 237–264, <http://dx.doi.org/10.1016/j.cpc.2015.02.014>.
- [74] P.E. Lapenna, G. Indelicato, R. Lamioni, F. Creta, Modeling the equations of state using a flamelet approach in LRE-like conditions, *Acta Astronaut.* 158 (2019) 460–469, <http://dx.doi.org/10.1016/j.actaastro.2018.07.025>.
- [75] E.L. Petersen, D.F. Davidson, R.K. Hanson, Kinetics modeling of shock-induced ignition in low-dilution CH₄/O₂ mixtures at high pressures and intermediate temperatures, *Combust. Flame* 117 (1) (1999) 272–290, [http://dx.doi.org/10.1016/S0010-2180\(98\)00111-4](http://dx.doi.org/10.1016/S0010-2180(98)00111-4).
- [76] G. Ribert, X. Petit, P. Domingo, High-pressure methane-oxygen flames. Analysis of sub-grid scale contributions in filtered equations of state, *J. Supercrit. Fluids* 121 (2017) 78–88, <http://dx.doi.org/10.1016/j.supflu.2016.08.020>.
- [77] A.Y. Klimenko, R.W. Bilger, Conditional moment closure for turbulent combustion, *Prog. Energy Combust. Sci.* 25 (6) (1999) 595–687, [http://dx.doi.org/10.1016/S0360-1285\(99\)00006-4](http://dx.doi.org/10.1016/S0360-1285(99)00006-4).
- [78] F. Nicoud, F. Ducros, Subgrid-scale stress modelling based on the square of the velocity gradient tensor, *Flow. Turbul. Combust.* 62 (1999) 183–200, <http://dx.doi.org/10.1023/A:1009959426001>.
- [79] K.A. Kemenov, H. Wang, S.B. Pope, Modelling effects of subgrid-scale mixture fraction variance in LES of a piloted diffusion flame, *Combust. Theory Model.* 16 (4) (2012) 611–638, <http://dx.doi.org/10.1080/13647830.2011.645881>.
- [80] P. Domingo, L. Vervisch, D. Veynante, Large-eddy simulation of a lifted methane jet flame in a vitiated coflow, *Combust. Flame* 152 (3) (2008) 415–432, <http://dx.doi.org/10.1016/j.combustflame.2007.09.002>.
- [81] S. Wang, H. Wang, K. Luo, J. Fan, The effects of differential diffusion on turbulent non-premixed flames LO₂/CH₄ under transcritical conditions using large-eddy simulation, *Energies* 16 (2023) <http://dx.doi.org/10.3390/en16031065>.
- [82] G.K. Giannakopoulos, A. Gatzoulis, C.E. Frouzakis, M. Matalon, A.G. Tomboulides, Consistent definitions of “Flame Displacement Speed” and “Markstein Length” for premixed flame propagation, *Combust. Flame* 162 (4) (2015) 1249–1264, <http://dx.doi.org/10.1016/j.combustflame.2014.10.015>.
- [83] F. Creta, R. Lamioni, P.E. Lapenna, G. Troiani, Interplay of Darrieus-Landau instability and weak turbulence in premixed flame propagation, *Phys. Rev. E* 94 (2016) 053102, <http://dx.doi.org/10.1103/PhysRevE.94.053102>.
- [84] D.O. Lignell, J.H. Chen, P.J. Smith, T. Lu, C.K. Law, The effect of flame structure on soot formation and transport in a temporally evolving nonpremixed ethylene jet flame, *Combust. Flame* 151 (1) (2007) 2–28, <http://dx.doi.org/10.1016/j.combustflame.2007.05.013>.
- [85] D.O. Lignell, J.H. Chen, P.J. Smith, Three-dimensional direct numerical simulation of soot formation and transport in a temporally evolving nonpremixed ethylene jet flame, *Combust. Flame* 155 (1) (2008) 316–333, <http://dx.doi.org/10.1016/j.combustflame.2008.05.020>.
- [86] T. Echekki, J.H. Chen, Analysis of the contribution of curvature to premixed flame propagation, *Combust. Flame* 118 (1) (1999) 308–311, [http://dx.doi.org/10.1016/S0010-2180\(99\)00006-1](http://dx.doi.org/10.1016/S0010-2180(99)00006-1).

- [87] L. Cutrone, P. De Palma, G. Pascasio, M. Napolitano, A RANS flamelet–progress-variable method for computing reacting flows of real-gas mixtures, *Comput. & Fluids* 39 (3) (2010) 485–498, <http://dx.doi.org/10.1016/j.compfluid.2009.10.001>.
- [88] J. Song, B. Sun, Coupled numerical simulation of combustion and regenerative cooling in LOX/Methane rocket engines, *Appl. Therm. Eng.* 106 (2016) 762–773, <http://dx.doi.org/10.1016/j.applthermaleng.2016.05.130>.
- [89] S. Priyadarshini, M.K. Das, A. De, R. Sinha, Numerical investigation of coaxial GCH₄/LOx combustion at supercritical pressures, *Combust. Sci. Technol.* 193 (11) (2021) 1973–1997, <http://dx.doi.org/10.1080/00102202.2020.1723009>.
- [90] A. Sharma, A. De, S.S. Kumar, Numerical investigation of supercritical combustion dynamics in a multi-element LOx–methane combustor using flamelet-generated manifold approach, *Phys. Fluids* 35 (11) (2023) 115125, <http://dx.doi.org/10.1063/5.0172100>.
- [91] H.G. Weller, G. Tabor, H. Jasak, C. Fureby, A tensorial approach to computational continuum mechanics using object-oriented techniques, *Comput. Phys.* 12 (6) (1998) 620–631, <http://dx.doi.org/10.1063/1.168744>.
- [92] G. Indelicato, F. Creta, Assessment of an algebraic equilibrium wall-function for supercritical flows, *Int. J. Heat Mass Transfer* 197 (2022) 123350, <http://dx.doi.org/10.1016/j.ijheatmasstransfer.2022.123350>.
- [93] G. Indelicato, A. Remiddi, P.E. Lapenna, F. Creta, N.P. Longmire, D.T. Banuti, Dataset of wall-resolved large-eddy simulations turbulent pseudobubbling in cryogenic hydrogen pipe flows, *J. Thermophys. Heat Transf.* 37 (1) (2023) 133–146, <http://dx.doi.org/10.2514/1.T6604>.
- [94] G. Indelicato, P.E. Lapenna, A. Remiddi, F. Creta, An efficient modeling framework for wall heat flux prediction in rocket combustion chambers using non adiabatic flamelets and wall-functions, *Int. J. Heat Mass Transfer* 169 (2021) 120913, <http://dx.doi.org/10.1016/j.ijheatmasstransfer.2021.120913>.
- [95] A. Remiddi, G. P.E. Lapenna, F. Creta, Efficient time-resolved thermal characterization of single and multi-injector rocket combustion chambers, *Proc. Combust. Inst.* 39 (4) (2023) 5043–5052, <http://dx.doi.org/10.1016/j.proci.2022.07.231>.
- [96] A. Remiddi, P.E. Lapenna, G. Indelicato, M. Valorani, M. Pizzarelli, F. Creta, Heat transfer in rocket combustion chambers firing plates: Role of injector confinement, *J. Propuls. Power* 39 (2) (2023) 176–189, <http://dx.doi.org/10.2514/1.B38847>.
- [97] D. Cavalieri, G. Indelicato, A. Remiddi, F. Creta, P.P. Ciottoli, P.E. Lapenna, A pressure-based numerical framework for highly stratified transcritical real-fluids simulations, in: *AIAA SCITECH Forum*, 2023, <http://dx.doi.org/10.2514/6.2023-1666>.
- [98] M.M. Jarczyk, M. Pfizner, Large eddy simulation of supercritical nitrogen jets, in: *50th AIAA Aerospace Sciences Meeting Including the New Horizons Forum and Aerospace Exposition*, 50th AIAA Aerospace Sciences Meeting, 2012, pp. 1–13, <http://dx.doi.org/10.2514/6.2012-1270>.
- [99] C. Traxinger, J. Zips, M. Banholzer, M. Pfizner, A pressure-based solution framework for sub- and supersonic flows considering real-gas effects and phase separation under engine-relevant conditions, *Comput. & Fluids* (ISSN: 0045-7930) 202 (2020) 104452, <http://dx.doi.org/10.1016/j.compfluid.2020.104452>.
- [100] D.N. Nguyen, C.S. Yoo, An OpenFOAM-based solver for modeling low mach number turbulent flows at high pressure with real-fluid effects, *Comput. Phys. Comm.* (ISSN: 0010-4655) 312 (2025) 109600, <http://dx.doi.org/10.1016/j.cpc.2025.109600>.
- [101] R.I. Issa, A.D. Gosman, A.P. Watkins, The computation of compressible and incompressible recirculating flows by a non-iterative implicit scheme, *J. Comput. Phys.* (ISSN: 0021-9991) 62 (1) (1986) 66–82, [http://dx.doi.org/10.1016/0021-9991\(86\)90100-2](http://dx.doi.org/10.1016/0021-9991(86)90100-2).
- [102] D. Papamoschou, A. Roshko, The compressible turbulent shear layer: an experimental study, *J. Fluid Mech.* 197 (1988) 453–477, <http://dx.doi.org/10.1017/S0022112088003325>.
- [103] H. Kröger, N. Kornev, Generation of divergence free synthetic inflow turbulence with arbitrary anisotropy, *Comput. & Fluids* 165 (2018) 78–88, <http://dx.doi.org/10.1016/j.compfluid.2018.01.018>.
- [104] I.B. Celik, Z.N. Cehreli, I. Yavuz, Index of resolution quality for large eddy simulations, *J. Fluids Eng.* 127 (5) (2005) 949–958, <http://dx.doi.org/10.1115/1.1990201>.
- [105] D.T. Banuti, K. Hannemann, The absence of a dense potential core in supercritical injection: A thermal break-up mechanism, *Phys. Fluids* 28 (3) (2016) 035103, <http://dx.doi.org/10.1063/1.4943038>.
- [106] T. Fiala, T. Sattelmayer, Assessment of existing and new modeling strategies for the simulation of OH* radiation in high-temperature flames, *CEAS Space J.* 8 (5) (2016) 47–58, <http://dx.doi.org/10.1007/s12567-015-0107-z>.